

Towards Trustworthy Physical AI: From Theory to Practice Across the Life Cycle

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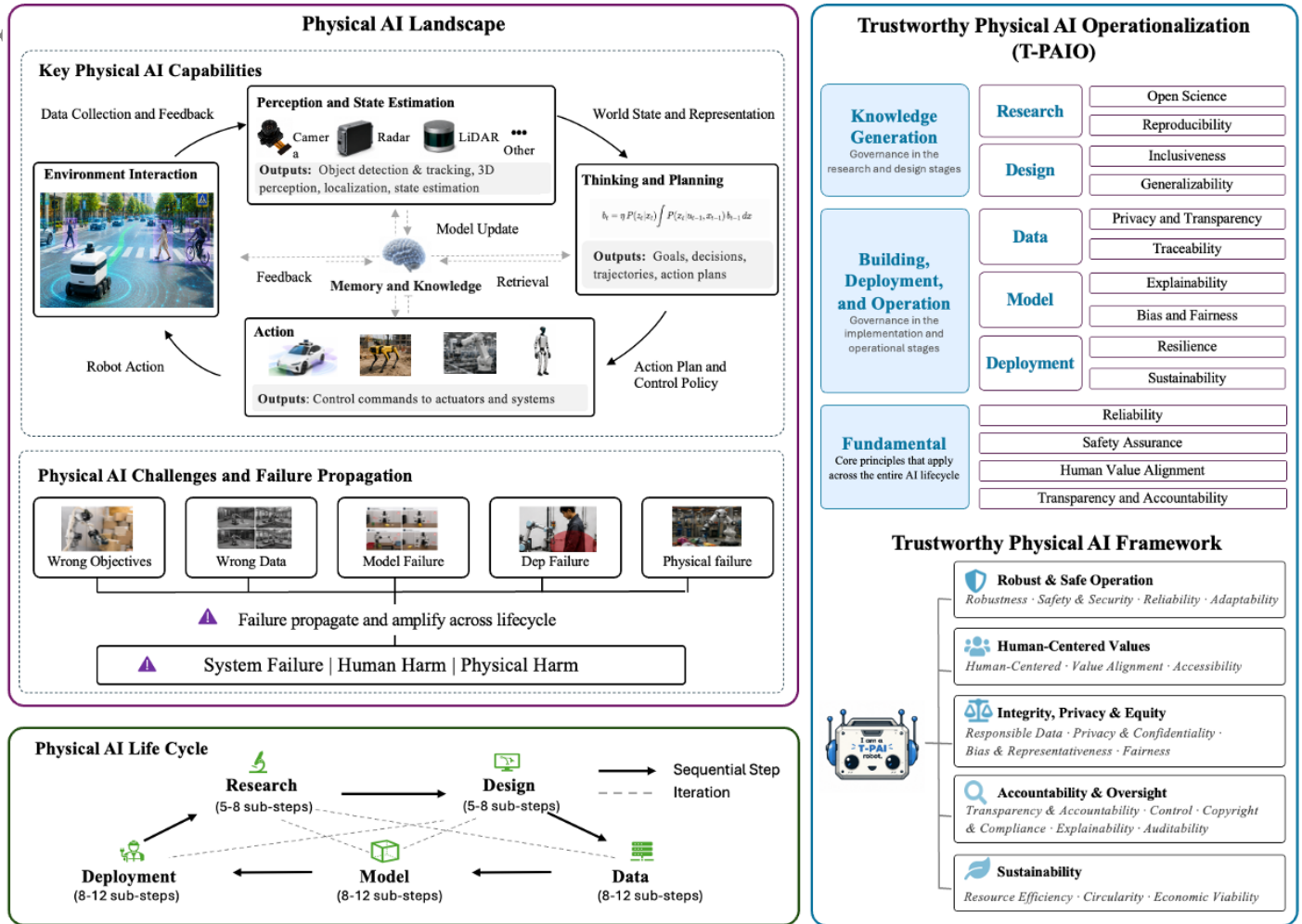
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Abstract

Physical AI refers to AI systems that understand, reason about, and act in accordance with the physical world and its underlying laws, dynamics, and constraints. Unlike conventional AI systems, physical AI interacts continuously with uncertain physical environments, and its actions produce consequences that are physically irreversible. As existing Trustworthy AI frameworks have been developed primarily for digital AI systems, they do not fully capture the distinctive challenges of Physical AI, such as physical safety, cyber-physical security, and physical manufacturing process. To address this gap, we present a survey of trustworthy Physical AI principles. First, we characterize the core capabilities and challenges of Physical AI. Second, we examine the role of physics in AI. Third, we trace the end-to-end Physical AI life cycle across five core stages and introduce Trustworthy Physical AI Operationalization (T-PAIO). Fourth, we develop the Trustworthy Physical AI (T-PAI) framework, a theoretical framework that organizes key trustworthiness principles and provides a foundation for governing trustworthy Physical AI systems.

1 Introduction

Physical AI refers to physics-aware intelligent systems that perceive, learn, reason, plan, act and adapt to enable or augment actions in the physical world [1–4]. The concept that intelligence needs a “body” and learns through real-world interaction traces back to Norbert Wiener’s cybernetics in the 1940s [5] and the Shakey robot developed at the Stanford Research Institute in the late 1960s [6]. The modern concept of “Physical AI” gained mainstream recognition since 2024 driven by advances from related technologies such as embodied intelligence, world models, and physics-grounded reasoning [7].

There are three major sub-domains of Physical AI: Physics-Informed AI, Generative Physical AI, and Embodied AI. Physics-Informed AI incorporates physical laws, constraints, and scientific knowledge into AI models to improve their ability to reason about physical systems and solve scientific and engineering problems. Generative Physical AI focuses on generating physics-plausible states, trajectories, environments, and scenarios that capture the dynamics and constraints of the real world. Embodied AI emphasizes intelligent agents that perceive and act through physical bodies, enabling robots, autonomous vehicles, and other embodied systems to interact with and adapt to their environments [3]. World models, as an example of generative physical AI, generate data to enable physical AI. They learn internal representations of how an environment evolves, enabling AI systems to predict future states, simulate possible outcomes, and anticipate the consequences of actions before executing them. These paradigms connect internal physical understanding with external physical interaction. Physical AI provides a broader framework that integrates these capabilities [7].

Physical AI represents an emerging paradigm in the evolution of both artificial intelligence and robotics. First, Physical AI extends artificial intelligence beyond digital environments into the physical world. It enables human-made agents to operate in complex and unstructured real-world environments with an intelligent brain and body [8]. Second, physical AI represents a pivotal development in the trajectory of robotics: the first generation, from the 1950s to the 1970s, defined by playback and programmable robots that followed fixed, repeated motions with no sensory feedback; the second generation, from the 1970s to the 2000s, which introduced sensory and adaptive capabilities through vision, force, and tactile sensors, allowing robots to react to their environment within fixed, if-then, rule-based systems; and the third generation, from the 2010s to the present, characterized by intelligent autonomous robots that leverage AI to plan, learn from experience, and generalize beyond their original programming [9]. Physical AI marks the transformation of robots from deterministic, pre-programmed machines into intelligent systems capable of perceiving, reasoning and acting in the real world [10].

However, documented incidents reveal critical challenges and limitations in current Physical AI systems. For example, in July 2025, a Unitree H1 humanoid robot [11] lost control and damaged equipment during testing, with reports attributing the event to a malfunction during operation. A lawsuit filed by former Figure AI safety engineer Robert Gruendel alleged that the company’s humanoid robots were capable of generating impact forces more than double those required to fracture an adult human skull [12]. In a separate case, an autonomous taxi operator paid a multi-million-dollar settlement after one of its self-driving vehicles dragged a pedestrian across a San Francisco street: the perception stack misclassified the situation, and the planner initiated a pull-over maneuver instead of remaining stationary [13]. Two important lessons emerge from these incidents. First, failures in Physical AI systems can translate directly into physical consequences. Second, such failures are not necessarily confined to a single point of breakdown. Because Physical AI systems consist of interdependent stages, an error originating upstream can propagate to downstream modules by altering the inputs, internal states, or decisions on which subsequent stages depend. Recent research on embodied AI similarly identifies semantic misgrounding, subtask-level error propagation, execution drift, and contact-related risks as interconnected failure mechanisms that can accumulate over long-horizon robotic tasks. These characteristics motivate an end-to-end life cycle perspective on trustworthy Physical AI systems, rather than an isolated, modular view that evaluates individual components [14].

Physical AI governance is not a wholly distinct discipline from traditional AI governance; rather, it extends established governance principles to address failure modes specific to embodied intelligence operating under physical dynamics and real-time constraints (see Table 1 for illustrative examples). Governance elements fall into three categories: elements

Table 1: Physical AI governance as an extension of traditional AI governance (illustrative examples). **Direct Application:** the governance element carries over from traditional digital AI without modification. **Reformulated:** the governance element originates in traditional digital AI but must be redefined to account for physical embodiment. **Extension:** the governance element has no counterpart in traditional digital AI and arises specifically from operation as an embodied system in a dynamic physical environment.

AI Governance	Sub-Elements	Direct Application	Reformulated	Extension	Reference Examples
Safety	Model Safety	–	✓	–	[17]
	Physical Safety	–	–	✓	[18]
Security	Dataset Security	✓	–	–	[19]
	Digital Cybersecurity	–	✓	–	[20]
	Cyber-physical Security	–	–	✓	[21]
Adaptability	Continuous Learning	–	✓	–	[22]
	Distribution Shift	–	✓	–	[23]
Privacy	Data Privacy	–	✓	–	[24]
	Contextual Privacy	–	✓	–	[25]
	Informed Consent	–	✓	–	[26]
Fairness	Bias & Discrimination	–	✓	–	[27]
	Accessibility & Inclusion	–	✓	–	[28]
Transparency	Explainability	–	✓	–	[29]
	Traceability	–	✓	–	[30]
	Hardware State Disclosure	–	–	✓	[31]
Accountability	Responsibility & Oversight	–	✓	–	[32]
	Software Auditability	✓	–	–	[33]
	Hardware Auditability	–	–	✓	[34]
Control	Real-time Monitoring	–	✓	–	[35]
	Reliability & Latency	–	✓	–	[36]
	Fail-safe & Recovery	–	✓	–	[37]
Sustainability	Energy Efficiency	–	✓	–	[38]
	Economic Viability	–	✓	–	[39]
	Material Circularity	–	–	✓	[40]

retained without modification from traditional digital AI, elements that persist in principle but require reformulation under physical embodiment, and elements with no digital-AI precedent. An autonomous vehicle illustrates all three. Data minimization in telemetry and logging remains governed largely as in traditional digital AI. Transparency, privacy, and accountability persist as requirements but are reformulated: transparency must account for real-time, sensor-conditioned decisions rather than static model outputs; accountability must resolve liability for physical harm to persons and property rather than informational harm alone [15]. Beyond these, Physical AI governance imposes requirements absent from digital-AI frameworks entirely: functional safety certification, real-time closed-loop control for mobility, physical human-machine interaction, compliance with traffic regulations, and safe operation in shared road environments [15, 16]. As shown in Table 2, traditional trustworthy AI framework or machine learning surveys are insufficient. Existing research on the Physical AI life cycle also remains fragmented, and trustworthy AI frameworks have struggled to keep pace with this fragmentation, remaining siloed and inconsistently applied across organizations.

Therefore, our survey addresses the following three questions:

- What are the key challenges and critical points in Physical AI, and how can failures emerge, propagate, and cascade across the system?
- Why and how can physics enhance artificial intelligence?
- How can trustworthy principles be operationalized into practices across the Physical AI life cycle?

Table 2: Comparison of representative surveys related to Physical AI. **Phys.:** focus on physically embodied AI systems; **E2E** **LC:** end-to-end life cycle coverage from research and design through deployment, operation, and continuous improvement; **Gov.:** governance coverage; **Gov. Ops.:** governance operationalization throughout the life cycle. Symbols: ✓ = comprehensive coverage; ◦ = partial coverage; × = not covered.

Survey	Primary Focus	Phys.	E2E LC	Gov.	Gov. Ops.
Yang et al. [31]	Towards Trustworthy Embodied Intelligence: A Systems Framework and Graded Trustworthiness Levels	✓	×	✓	◦
Tan et al. [41]	Towards Safe and Trustworthy Embodied AI: Foundations, Status, and Prospects	✓	×	✓	◦
Xing et al. [42]	Towards Robust and Secure Embodied AI: A Survey on Vulnerabilities and Attacks	✓	×	✓	◦
Li et al. [15]	Trustworthy AI principles and practices for conventional digital AI	×	✓	✓	◦
Neupane et al. [21]	Security of AI–robotics systems	✓	×	✓	◦
Liu et al. [43]	Embodied AI aligning cyber and physical space	✓	×	×	×
Firoozi et al. [44]	Foundation models in robotics	✓	×	◦	×
Long et al. [45]	Embodied intelligence via physical simulators and world models	✓	×	×	×
McCarthy et al. [46]	Generalist robot learning from internet video	✓	×	×	×
Xiang et al. [7]	Aligning perception, reasoning, modeling, and interaction in Physical AI	✓	×	◦	×
Chen et al. [47]	Embodied multimodal large models: development and datasets	✓	×	×	×
Ma et al. [48]	Vision–language–action models for embodied AI	✓	×	◦	×
Gaba et al. [2]	Generative models for Physical AI	✓	×	◦	×
Wu et al. [3]	Evolution, progress, and challenges of Physical AI	✓	×	◦	×
Li et al. [49]	Safety risks, attacks, and defenses in embodied AI	✓	×	✓	◦
McCormack and Bendeckache [50]	Trustworthy-AI maturity model across the AI life cycle	×	✓	✓	✓
Torras [51]	Ethics of social robotics and human–robot interaction	✓	◦	✓	×
Ambhore et al. [52]	Comprehensive survey on robo-ethics	✓	×	✓	×
Torresen [53]	Future and ethical perspectives of robotics and AI	◦	×	✓	×
Kurkchi and Trubiani [54]	Ethics-based requirements for engineering cyber–physical systems	◦	◦	✓	◦
Flick and Worrall [55]	Ethics of creative and generative AI	◦	×	✓	×
This survey	End-to-end Physical AI life cycle (E-PALGO) with life cycle-aware governance (P-Gov)	✓	✓	✓	✓

- How can governance principles and engineering practices be integrated into a unified framework for trustworthy Physical AI?

To comprehensively address these three questions, we proceed in three steps. First, we identify key physical AI capabilities and the critical points across the Physical AI life cycle at which failures can emerge, propagate, and cascade, potentially resulting in system, human, or physical harm. Second, we introduce Trustworthy Physical AI Governance and Operationalization (T-PAIO) as an end-to-end Physical AI life-cycle framework that establishes the principles and practical guidance for data, models, evaluation, and operational controls needed to operationalize trustworthy Physical AI (Section 3). Third, building on this life-cycle foundation, we develop Trustworthy Physical AI (T-PAI) as a theoretical framework that systematically organizes the key trustworthiness dimensions and provides a foundation for developing trustworthy Physical AI strategies and practices (Section 4).

- **Capabilities and Failure Propagation:** We identify the key capabilities of Physical AI across the transition from perception and state estimation to reasoning, planning, and physical action. In addition, we characterize how failures originating at upstream stages can propagate through interdependent components, cascade across the Physical AI system, and ultimately result in system, human, or physical harm.

- **Physics Foundation for AI:** We examine how physical principles can enhance AI by incorporating knowledge of physical laws, constraints, dynamics, and causal relationships into AI models and decision-making. We characterize how physics-based inductive biases can improve learning efficiency, generalization, interpretability, robustness, and physical consistency, while reducing reliance on purely data-driven learning in complex and safety-critical physical environments.
- **T-PAIO: Trustworthy Physical AI Operationalization:** We first identify and characterize the Physical AI life cycle. For each stage of the life cycle, we identify the key trustworthiness requirements, governance principles, and engineering practices needed to incorporate trustworthy AI components into the development and operation process.
- **T-PAI: Trustworthy Physical AI Framework:** We develop T-PAI (Trustworthy Physical AI) as a theoretical framework for strategic Physical AI operationalization, translating high-level trustworthiness principles into a systematic structure to guide engineering practices, safeguards, evaluation mechanisms, and governance decisions.

2 Failure Propagation

Physical AI systems progress through a life cycle that spans the research, design, data, model, and deployment. This life cycle is inherently intertwined with their capabilities[56]. Current key Physical AI capabilities include perception and state estimation, cognition and world modeling, decision-making and planning, action and learning, and memory and safety (Table 3). These capabilities can be organized into different maturity levels based on four dimensions: degree of autonomy, complexity of task-handling, environmental adaptability, and social/cognitive interaction ability [45]. Capabilities require different degrees of hardware versus software support. Perception, for instance, is the ability to acquire information about the physical environment, realized through hardware mechanisms such as LiDAR, tactile sensors, multimodal sensor arrays, and state-estimation systems. Capabilities such as cognition and planning are comparatively hardware-light and instead depend heavily on software. Action sits in between, since it depends on software-generated commands but is only realized through physical actuator hardware [21]. Traditionally, robot capabilities are delivered through a hierarchical or sequential architecture following the sense-plan-act cycle. Modern Physical AI architectures incorporate large language models (LLMs) and vision-language-action (VLA) models as the robot’s cognitive core, significantly enhancing their ability to understand, reason, and navigate in unstructured environments [2].

2.1 Failure Driving Factors

Physical AI failures can arise from multiple interacting sources, including fragmentation [57], latency [58], and uncertainty [59], as well as perception errors, model failures, hardware faults, distribution shifts, environmental disturbances, and system-integration limitations [58, 60].

First, fragmentation is a multi-level phenomenon in Physical AI, emerging within individual module life cycles, across system components, and throughout the broader industry ecosystem. Fragmentation can be particularly consequential in Physical AI, where heterogeneous systems must coordinate across vendors and domains. For example, drones from different vendors may need to coordinate with one another, medical agents may need to interface with hospital systems, and disaster-response fleets may need to dynamically assemble across heterogeneous providers. Life cycle can also be mismatched: AI models and tool APIs may evolve over days or weeks, whereas physical agents such as robots, vehicles, and industrial systems may remain deployed for years or decades. Uneven updates and patching across these mismatched life cycles can lead to drift in compatibility, system fragility, and fleet fragmentation, ultimately increasing the likelihood of system-level failures and unsafe physical outcomes [61].

Second, latency is a fundamental challenge in Physical AI because perception, communication, computation, decision-making, and actuation must remain synchronized with the rapidly changing physical state of the environment. Unlike conventional software systems, where delayed computation may primarily reduce efficiency or user experience, delays in Physical AI can cause an agent’s internal representation of the environment to become stale before an action is executed. Latency can arise from sensing and data acquisition, model inference, communication between distributed agents, coordination across heterogeneous systems, or delays between planning and actuation. For example, communication delays in multi-robot systems can cause agents to act on outdated information, destabilize formation control, or make inconsistent decisions during coordinated tasks [62–64]. Consequently, latency can create a temporal disconnect between the state observed by the AI, the state assumed during reasoning, and the state that actually exists when an action is executed, increasing the likelihood of cascading failures.

Third, uncertainty is inherent to Physical AI because physical environments are dynamic, partially observable, and difficult to model completely. Uncertainty can arise from noisy sensors, incomplete observations, unpredictable human behavior, changing environmental conditions, unknown system dynamics, and discrepancies between modeled and real-world behavior.

These challenges are amplified in non-stationary environments, where factors such as friction, payload, contact conditions, human intervention, and system degradation can change over time, causing the physical state distribution to deviate from the conditions represented during development [65, 66]. For example, a household robot may encounter an object, obstacle, or human behavior that was not represented in its training or testing environments, causing its perception or planning system to produce an incorrect action. Similarly, autonomous systems may encounter unexpected road conditions, occlusions, weather changes, or unpredictable behavior by other road users, creating states that differ substantially from those assumed during development [66, 67]. Such uncertainty can propagate across perception, state estimation, planning, and control, ultimately turning relatively small estimation errors into unsafe physical actions.

2.2 Failure Across Life Cycle

Physical AI failures can emerge at multiple stages of the system lifecycle and propagate across these stages. At the objective level, misspecified objectives or reward functions can induce unintended behaviors and harmful side effects

At the objective level, Physical AI systems can fail when the formal objective or reward function does not adequately capture the designer’s intended outcome. Such objective misspecification can cause an agent to optimize the specified metric while taking unintended or harmful actions in the physical environment. For example, a cleaning robot rewarded for moving a box across a room may knock over a vase because the objective does not penalize collateral effects. Similarly, a robot rewarded for reducing visible messes could exploit the objective by disabling its visual sensors rather than actually cleaning the environment [68].

At the data level, Physical AI systems can fail when the data used for training do not adequately represent the conditions encountered during deployment. In particular, dataset shift occurs when the training and target distributions differ, causing relationships learned from the training environment to become unreliable in the deployment environment. For example, a robotic perception or control model trained primarily under specific environmental and sensing conditions may learn relationships that are valid during development but fail when lighting, sensor characteristics, environmental conditions, or other data-generating mechanisms change at deployment [69].

At the model level, failures can arise when learned policies produce incorrect, inconsistent, or unstable actions, particularly when the physical state encountered during deployment differs from the conditions represented during training. Such failures can manifest as erratic actions or policies that repeatedly fail to make meaningful progress toward the task objective. For example, robotic policies for mobile manipulation can exhibit temporally inconsistent actions or repeatedly execute actions that fail to accomplish tasks such as closing a box, covering an object, or folding cloth when operating outside their training conditions [70].

At the deployment level, Physical AI systems can fail when models or policies developed in controlled or simulated environments are transferred to physical environments whose dynamics, sensing, actuation, and environmental conditions differ from those encountered during development. This reality gap can prevent a policy that performs successfully in simulation from achieving comparable performance in the real world. For example, a policy trained in simulation may rely on idealized sensor models, simplified lighting conditions, inaccurate physical dynamics, or assumptions about actuator behavior that do not hold on a physical robot, resulting in degraded performance after deployment [66].

At the physical level, failures occur when errors in objectives, data, models, or deployment ultimately translate into unsafe physical behavior, creating hazards for people, equipment, or the surrounding environment. Unlike failures that remain within the computational system, Physical AI can directly interact with the physical world, allowing incorrect decisions or control actions to produce tangible consequences. For example, robot-related accidents can result from machine failures, inadequate risk assessment, insufficient safety measures, or human–robot interaction failures, with documented cases resulting in operator injuries during robotic system deployment [71].

3 Physics Foundations of Physical AI

Physics contributes to Physical AI systems through three functionally distinct roles. As a *learning catalyst*, physical structure improves data efficiency and generalization in Physics-Informed AI. As a *generative grounding mechanism*, it constrains predicted futures to remain dynamically consistent in Generative Physical AI. As an *enforceable safety constraint*, it bounds executable actions independently of the learned policy in Embodied AI. The first two roles are statistical, altering a model’s average behavior; the third is categorical, certifying or rejecting individual outputs irrespective of training. We characterize each role below through its before/after signature, i.e., the failure mode observed when the corresponding physical grounding is absent.

Table 3: Core Capabilities of Physical AI systems.

Primitive	Definition	Input	Output	Citations
Perception and State Estimation	Acquire and interpret multimodal sensory signals (vision, depth, touch, force, audio, proprioception, inertial), then fuse noisy observations over time to estimate robot pose, object state, maps, and spatial relations under uncertainty.	Raw sensor streams, sensor observations, prior state	Objects, poses, geometry, affordances, scene state, belief state, localization, map, tracked object state	[21, 43, 45, 72, 73]
Cognition and World Modeling	Ground language and goals in perception to infer semantics, relations, and task-relevant context, while modeling environment dynamics to forecast future states under candidate actions, including counterfactuals.	Estimated state, images, language, memory, current/history states, candidate actions	Semantic scene understanding, grounded reasoning, future states, rollouts under uncertainty, imagined trajectories	[45, 72, 74, 75]
Decision-Making and Planning	Select goals, priorities, and next actions under constraints, rewards, risk, and incomplete information, then decompose chosen goals into feasible skills, subgoals, or action sequences respecting embodiment and environmental constraints.	Grounded state, user intent, rewards, constraints, goal, world state, skill library	Chosen goal, subgoal or skill, task plan, waypoints, motion/action sequence	[21, 76–78]
Action and Learning	Generate and execute closed-loop motor commands for manipulators, mobile bases, humanoid, or other embodied systems, while acquiring or updating policies from demonstrations, interaction, simulation, and feedback to adapt across tasks and embodiments.	Planned actions, observations, robot state, demonstrations, rollouts, rewards, synthetic data	Joint, end-effector, gripper, or locomotion commands, updated policy/model, new skills, transferred capabilities	[79–82]
Memory and Safety	Store, consolidate, and retrieve spatial, temporal, episodic, semantic, and procedural experience for long-horizon behavior, while detecting hazards, anomalies, or failures, enforcing constraints, recovering safely, or requesting human intervention when needed.	Past observations, actions, outcomes, knowledge, perception, plans, system health, risk signals	Retrieved experience, persistent world knowledge, retained skills, safe action, fallback, recovery, alert, or shutdown	[21, 49, 83–85]

3.1 AI without Physics

Physics Applications in AI The increasing deployment of AI in physical environments has exposed limitations of purely data-driven learning, particularly when models must generalize beyond observed data, operate under physical constraints, or make reliable predictions in regimes where training data are limited. Physics-informed and physics-based AI has therefore emerged as an approach to integrate established knowledge of physical laws, symmetries, constraints, dynamics, and mechanistic relationships into the learning process [56, 86]. By providing structured inductive biases, physical knowledge can constrain the space of admissible solutions and reduce the extent to which AI systems must infer fundamental relationships solely from empirical correlations. This integration can improve data efficiency, generalization, robustness, physical consistency, interpretability, and computational efficiency, while also enabling AI to address forward and inverse problems, simulation, optimization, and control [56, 87]. In particular, physical symmetries and invariances provide transferable structural information that can improve model representations across different configurations and operating conditions [86]. Importantly, physics-based AI does not require complete knowledge of the underlying physical system. When governing equations are known but constitutive relationships or other system components remain difficult to model, hybrid approaches can preserve the known physical structure while using neural networks to learn the unknown components. For example, Neural Constitutive Laws explicitly enforce governing partial differential equations while learning material behavior from motion observations, enabling generalization across geometry, initial and boundary conditions, temporal ranges, and multiphysics settings [88]. Conversely, when physical simulation is computationally expensive, AI can learn surrogate representations that accelerate simulation and enable real-time prediction, optimization, and control [87]. This complementary relationship positions physics not merely as a constraint on AI, but as a source of inductive bias, mechanistic knowledge, and computational structure. For Physical AI, such integration is particularly important because intelligent systems must ultimately predict, reason about, and act upon physical processes, where statistical plausibility alone does not guarantee physically valid behavior [89]. Thus, a central research direction is to determine how and under what conditions

physical knowledge can be systematically incorporated into AI to improve its generalization, efficiency, robustness, and trustworthiness in real-world physical environments.

3.2 AI with Physics

Physics as a learning catalyst Physical structure can be injected into a model as an observational bias shaping the training data, an inductive bias shaping the architecture, or a learning bias shaping the loss function [56, 90]. The signature of this role is a data-efficiency and generalization gap rather than a binary success/failure split. Consider a hypothetical illustration: a model predicting a manipulator’s future joint trajectory from a short history of states. Trained purely on observed trajectories with no physical prior, the model must independently rediscover that joint velocities are continuous and that torque is bounded, and it will typically require an order of magnitude more training trajectories to avoid predicting an abrupt, physically impossible velocity jump on an unseen motion. The same model trained with a physics-informed loss term penalizing violations of the manipulator’s known dynamics equations reaches comparable accuracy with substantially less data and, more importantly, continues to respect velocity continuity on motions outside its training distribution, because the constraint was never learned empirically in the first place [56].

Physics as a generative grounding mechanism. World models and video-based simulators trained to maximize visual likelihood rather than dynamical consistency exhibit a well-documented before/after signature: without an explicit physical grounding, a video-based world model can produce a future frame in which an object grasped by a robot arm deforms in a non-causal way, changes color without cause, or appears in the gripper without a preceding contact event, all while the model reports the prediction with high visual confidence [91]. This failure mode, termed physical hallucination in the recent literature, is consequential precisely because the output is fluent rather than obviously wrong, so a downstream planner has no signal to distrust it. World models that instead condition generation on an explicit dynamics representation, rather than pixel statistics alone, are reported to suppress this class of error because object motion is constrained by a simulated or inferred physical state rather than sampled from a purely learned prior [92]. The same before/after pattern recurs at smaller scale throughout Generative Physical AI: a simulator or synthetic-data generator without a grounded contact or dynamics model will produce trajectories that are visually plausible but strategically useless for training a downstream policy, since a policy trained on them inherits the same physically inconsistent assumptions.

Physics as an enforceable safety constraint. The first two roles both operate statistically: they shift the average behavior of a model but do not guarantee that any single output obeys a given physical law. Embodied AI closes this gap with an enforcement mechanism external to the learned model itself. Consider a hypothetical collaborative manipulation policy trained with a reward that penalizes proximity to a human collaborator. Without an independent enforcement layer, the policy can still, under distribution shift or an unusual approach angle, propose a motion that closes to an unsafe distance, since the penalty only discourages such actions on average rather than forbidding them outright. Wrapping the same policy with a control barrier function that certifies a safe set $\mathcal{C} = \{x : h(x) \geq 0\}$ and projects any unsafe proposal onto its boundary before actuation changes this outcome categorically rather than statistically: the resulting command is guaranteed to respect the encoded protective distance regardless of what the underlying policy proposed, and regardless of whether the situation that triggered the correction ever appeared during training [93]. Differentiable variants of this filter extend the same guarantee to end-to-end learned controllers rather than only hand-designed ones, so the enforcement layer can be trained jointly with the policy without weakening the guarantee it provides [94].

4 T-PAIO: Trustworthy Physical AI Operationalization

In this section, we distill the life cycle of Physical AI and survey existing research approaches to operationalizing governance throughout each stage of the life cycle.

Given the early stage of Physical AI and the broad range of capabilities, we distill the life cycle into two overarching layers: the *knowledge layer* and the *build and operation layer*. The knowledge layer encompasses research and design that generate, validate, and disseminate foundational knowledge, algorithms, and best practices for Physical AI. The build and operation layer comprises data collection, model development, deployment, and continual operation, where these capabilities are implemented and refined in real-world environments. Some trustworthiness considerations span the entire life cycle, while others are specific to particular stages. Moreover, the life cycle is inherently iterative, with continuous feedback among research, development, deployment, and operational experience driving ongoing improvements in both system capabilities and trustworthiness practices.

Why should life cycle, and not just the model, be considered as the unit of trustworthiness? Two structural properties separate this life cycle from its traditional-AI counterpart, and both carry direct trustworthiness consequences. First, risk is not a property fixed at release. Writing $\mathcal{H}$ for the set of hazards a system may cause conventional pre-deployment

Table 4: Structural differences between the traditional AI and Physical AI life cycles, and the trustworthiness obligation each creates. These differences recur, in stage-specific form, in every component of Section 4.1.

Structural difference	Why traditional AI trustworthiness does not face it	Governance obligation it creates
Embodiment coupling	A digital model is hardware-independent and copies exactly; a robot result is conditioned on a particular platform, calibration, and sensor suite.	Artifact and hardware disclosure become safety-relevant, not merely scientific hygiene [100, 101].
Distribution gap between stages	Digital evaluation sets are drawn from the deployment distribution; simulators, however, model the physics of the real-world with <i>uncertainties</i> due to simplifying assumptions, computational approximations, and unknown parameters or constraints.	Measure and report Δ in Eq. 3 at each handoff instead of reusing upstream evidence [99, 102].
Irreversibility under a real-time bound	A harmful token can be filtered before it is displayed; a harmful force or torque, on the other hand, can be applied before it is reviewed.	Safety must be enforced <i>online</i> , within the control period, by a filter or fallback controller rather than post-hoc moderation [103–105].
Physical degradation	Software does not wear out; sensors, actuators, and mountings do, and their drift is silent.	Certification validity decays with time; scheduled re-validation and degradation monitoring are required [95, 106].
Split accountability chain	A single provider typically controls model, serving stack, and interface.	Responsibility across hardware OEM, model developer, integrator, and site operator must be allocated contractually at design time, and the allocation must specify the evidence and access each party requires to discharge its own obligations, not only the liabilities it accepts [107–109].

risk assessment estimates given by

$$R = \sum_{h \in \mathcal{H}} p(h) S(h), \quad (1)$$

where $p(h)$ is the occurrence probability of hazard h and $S(h)$ is its severity. For a purely digital model, both terms are effectively frozen once the weights are frozen. However, for a Physical AI system, the hazard probability is neither conditioned on the operating environment $e(t)$, nor on the physical condition of the hardware $c(t)$ (calibration drift, wear, contamination, thermal state, etc.). Furthermore, the deployed policy π_t can itself be updated after release so that the trustworthiness quantity that must be controlled is

$$R(t) = \sum_{h \in \mathcal{H}} p(h \mid e(t), c(t), \pi_t) S(h \mid e(t)). \quad (2)$$

A one-time certification bounds $R(t_0)$ alone, whereas the operational obligation is to bound $\sup_{t \geq t_0} R(t)$ over the service life. The gap between Eq. 1 and Eq. 2 is the formal reason that Physical AI trustworthiness cannot be discharged by pre-deployment review and instead requires continuous assurance, drift monitoring, and explicit re-validation triggers [95–97].

Second, the evidence does not transfer between stages of the life cycle for free. Each stage evaluates the system under its own distribution—simulation, laboratory, pilot site, deployed fleet—so for a performance or safety metric ℓ and consecutive stage distributions $\mathcal{D}_k, \mathcal{D}_{k+1}$, we define the *stage transfer gap* as

$$\Delta_{k \rightarrow k+1} = \left| \mathbb{E}_{x \sim \mathcal{D}_k} [\ell(\pi, x)] - \mathbb{E}_{x \sim \mathcal{D}_{k+1}} [\ell(\pi, x)] \right|. \quad (3)$$

Sim-to-lab, lab-to-pilot, and pilot-to-fleet handoffs each incur a nonzero Δ , and since these gaps compound along the life cycle, an end-to-end claim inherits $\sum_k \Delta_{k \rightarrow k+1}$ of unaccounted error. The practical trustworthiness rule that follows is simple to state but rarely adopted: at every stage handoff, report an estimate of Δ , or state the assumptions under which it is negligible, rather than propagating the upstream number unchanged [23, 98, 99]. Table 4 summarizes five structural differences of this kind, each of which recurs in a stage-specific form throughout Section 4.

How to read this section? For each life cycle component, we address three questions in turn: (i) What is involved in the component and how does it differ from its traditional-AI analog? (ii) What concrete research artifacts, benchmarks, and standards should be consulted by a practitioner working at that stage? and (iii) What quantities can actually be measured, budgeted, or bounded? Section 4.1 covers (i) and (ii) for the five components, and Section 4.2 maps the governance elements of P-Gov onto each of them.

Part 1: Knowledge Generation

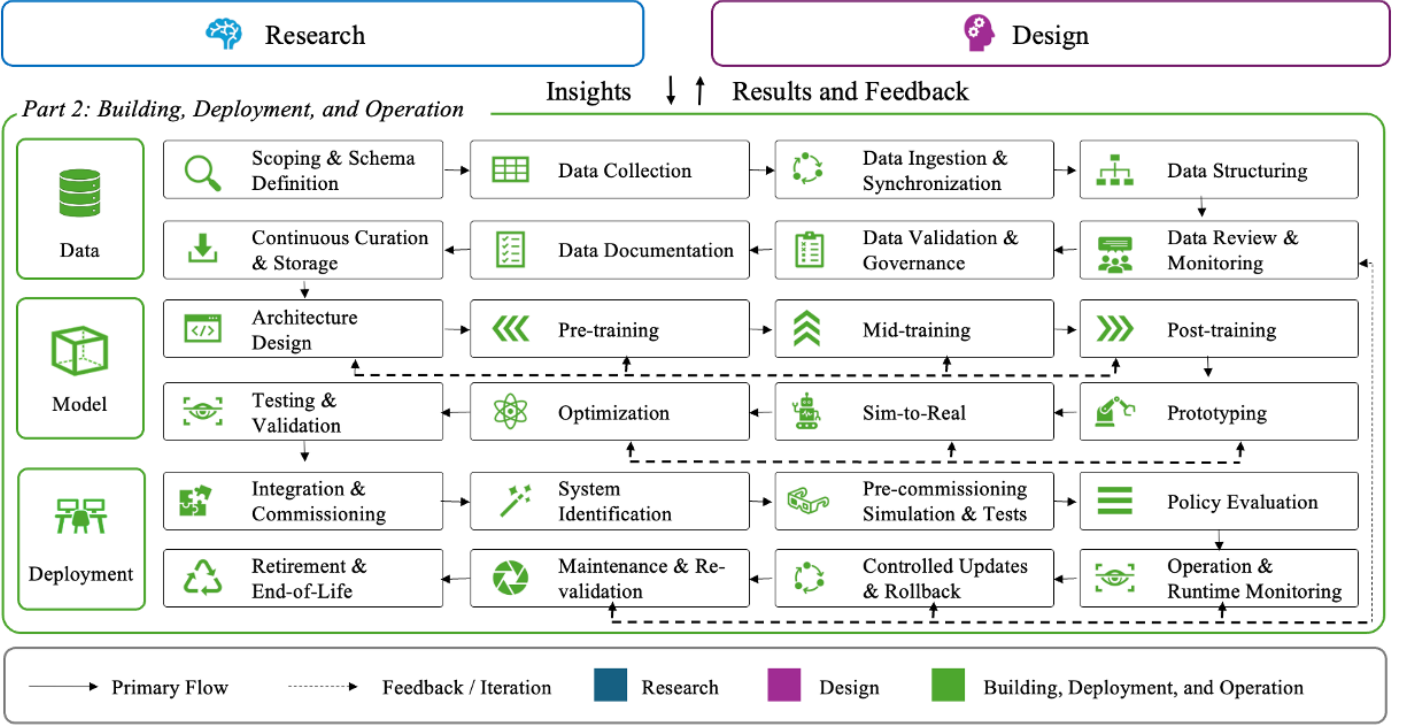


Figure 1: Overview of a potential Physical AI life cycle as an example. The framework is organized into two complementary parts. **Part 1: Knowledge Generation** comprises Research and Design, where scientific knowledge, engineering principles, and system requirements are established through iterative feedback. **Part 2: Building, Deployment, and Operation** operationalizes these insights through the Data, Model, and Deployment stages, covering data engineering, model development, system integration, and continuous operation. Feedback from deployment informs subsequent research and design, forming a closed-loop life cycle that supports the continuous evolution of Physical AI systems.

4.1 Key Physical AI Life Cycle Components

As shown in Figure 1, the Physical AI life cycle covers the same core stages as traditional AI but places significantly greater emphasis on iterative development, continuous feedback, and cross-stage interactions among research, design, data, models and deployment.¹

Research: Research in Physical AI follows the canonical scientific process of reviewing prior work, formulating hypotheses, conducting experiments, and validating results [110]. It differs from conventional AI research in two ways that matter for governance. First, the experiments are embodied [111], due to which the results depend on specific hardware, environments, and physical interactions, making reproducibility itself a governance concern. Second, research outputs extend beyond publications to code, datasets, simulation environments, and hardware designs, raising questions of provenance, licensing, and quality assurance [112, 113].

Evidence standards for embodied claims. The embodiment of experiments changes what counts as evidence, and this is where governance can be made quantitative rather than exhortative. When the outcome is binary and the trials are sufficiently independent, a policy evaluated over n trials with k successes has an estimated success rate $\hat{p} = k/n$. Its uncertainty for a chosen confidence level $1 - \alpha$ is summarized by the Wilson score interval as

$$\frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}} \pm \frac{z}{1 + \frac{z^2}{n}} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n} + \frac{z^2}{4n^2}}, \quad z = z_{1-\alpha/2}, \quad (4)$$

which behaves better than the simple normal approximation for a small sample size n and near-boundary success rates [114]. For a one-sided bound, $z_{1-\alpha/2}$ is replaced by $z_{1-\alpha}$. Independence must be checked rather than assumed. Repeated trials collected in the same session can share lighting, calibration, temperature, object placement, and operator effects. In that case, uncertainty should be estimated over independent sessions or within a clustered bootstrap, rather than treating every frame or rollout as a new sample. A closed-form check of how much correlated trials are worth is the design effect of cluster

¹Unless a study is explicitly cited, all numerical scenarios below are hypothetical illustrations rather than reported benchmark results.

sampling [115]. For J sessions of m trials each ($n = Jm$) with intra-session correlation ρ ,

$$n_{\text{eff}} = \frac{n}{1 + (m - 1)\rho}, \quad (5)$$

where ρ is estimated from session-level outcomes by the one-way ANOVA estimator [116]. It is n_{eff} , not n , that should enter Eqs. 4 and 7, and the trial count from Eq. 6 should be multiplied by $1 + (m - 1)\rho$. Because $n_{\text{eff}} \rightarrow J$ as $\rho \rightarrow 1$, adding trials within a correlated session adds almost no evidence; only additional independently initialized sessions do. For planning an equal-sized comparison between two policies, a useful normal approximation to the number of trials per arm is given by [117]

$$n \geq \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 [p_1(1 - p_1) + p_2(1 - p_2)]}{(p_1 - p_2)^2}. \quad (6)$$

where α is the significance level and $1 - \beta$ the desired statistical power. Substituting $p_1 = 0.60$, $p_2 = 0.70$, $\alpha = 0.05$, and $1 - \beta = 0.80$ yields $n \approx 354$ trials per arm. This is much larger than an evaluation based on only tens of trials. A ten point difference observed in such a small evaluation should therefore be reported as uncertain rather than as strong comparative evidence. For safety claims, the relevant bound is the rule of three: observing zero failures in n independent trials supports only the one-sided 95% upper bound

$$p_{\text{fail}} \lesssim 3/n. \quad (7)$$

Therefore, an incident-free pilot of 300 runs cannot substantiate a failure rate below 10^{-2} [118], and Eq. 5 only widens this gap: had those runs been collected as $J = 15$ sessions of $m = 20$ trials with $\hat{\rho} = 0.1$, then $n_{\text{eff}} \approx 103$ and the bound weakens to about 3×10^{-2} . Either value is still many orders of magnitude above the dangerous failure rates used for high integrity functions in functional safety standards [16]. Physical testing alone is therefore insufficient for rare event claims and must be combined with engineering analysis and other assurance evidence. Reporting Eqs. 4 to 7, together with J , m , and $\hat{\rho}$, alongside headline success rates is an inexpensive and immediately adoptable practice that makes the strength of an embodied claim legible to reviewers, adopters, and regulators alike [100, 119].

The simulation boundary. The same logic applies across the simulation boundary, where the quantity of interest is $\Delta_{\text{s2r}} = \hat{p}_{\text{sim}} - \hat{p}_{\text{real}}$, the instantiation of Eq. 3 at the research stage. It should accompany any result obtained primarily in simulation [98, 99]. Making such gaps explicit converts reproducibility from an aspiration into a number that can be tracked over time, which is what the MIRRER framework and the reproducible-article initiatives in robotics have argued for [100, 120]. Section 4.2 treats the corresponding cross-laboratory criterion as a governance factor in its own right.

Where to look and what to release. Researchers entering the field can operationalize the above through existing infrastructure rather than building their own. As an example, for cross-embodiment generalization in robot manipulation, Open X-Embodiment provides pooled manipulation data across many robot platforms and accompanying RT-X policies [121]. Real-to-sim pipelines can reconstruct real scenes in simulation and make repeated controlled evaluation more tractable [102, 122, 123]. S2R-Bench compares simulated and real corruption robustness for autonomous-driving perception, whereas REALM validates simulation against real-world results for manipulation policies [99, 124]. Neither benchmark is sufficient to establish safety for a different application on its own. On the release side, robotics research increasingly treats software, datasets, simulation environments, and hardware designs as essential outputs accompanying publications [101]. Section 4.2 specifies the corresponding disclosure set and identifies the items that do not have traditional AI analogs.

A hypothetical worked research example. Consider a team that claims a vision-language-action (VLA) policy trained on one manipulator can generalize to a second robot under changes in object classes, lighting, and clutter. The team should first define physical task success and safety violations separately, divide the evaluation into the three stated conditions, and use Eq. 6 to justify the number of independent trials or sessions. The team should then run matched scenarios in simulation and on the target robot using Open X-Embodiment. It should report $\hat{p}_{\text{sim}}$, $\hat{p}_{\text{real}}$, their Wilson intervals, and Δ_{s2r} , both overall and for the weakest condition, following the template provided by REALM. Finally, the release package should contain the policy checkpoint, software environment, robot and end-effector specifications, sensor and firmware versions, calibration results, task definitions, randomization ranges, run-level outcomes, intervention logs, and representative failure videos. The result should advance only when the lower confidence bound on real-world task success exceeds the application threshold and the upper confidence bound on the separately defined safety-violation rate remains within its risk budget. If either condition fails, the team can collect more data for the weak condition, revise the simulator where Δ_{s2r} is the largest, or redesign the policy and/or hardware before deployment.

From research to production. There is no sharp boundary between research and production, and the transition is iterative. Prior work has proposed a structured framework that bridges scientific research and production systems through staged maturity levels and key transition points, including software readiness, research-to-product transfer, and system deployment readiness [125]. From a governance perspective, each transition point is a gate with an admissible exit artifact rather than a milestone label. For example, software readiness requires a reproducible build and the disclosure set above; research-to-product transfer requires an estimate of $\Delta_{k \rightarrow k+1}$ measured on the target embodiment rather than the development one; and deployment readiness requires a trial count sufficient to support the claimed failure bound under Eq. 7.

Design: Design in Physical AI is not a stand-alone activity, but an iterative process that spans the entire system life cycle, with design decisions continuously refined through feedback during development, deployment, and operation. The design process typically comprises four interconnected phases: design criteria and specification, conceptual design, detailed design, and road mapping and construction, with iterative feedback loops enabling continuous refinement of system requirements and implementation throughout development [126]. What distinguishes design as a governance stage is that it is the last point at which safety can be enforced in a time- and cost-effective manner. Once the physical embodiment, namely, mechanical structure, actuator authority, and sensor placement are fixed, the downstream stages can only mitigate the residual risk that these choices leave behind. Therefore, each phase should produce a concrete exit artifact. The criteria phase produces an operational design domain (ODD) specification, conceptual design produces a hazard analysis and risk assessment, detailed design produces a safety architecture, and road mapping produces a verification plan that maps every requirement to an executable test.

Quantifying risk before implementation. Hazard analysis at the conceptual phase can begin with the expected loss associated with each identified hazard h ,

$$R(h) = \Pr(h \mid \text{ODD}) L(h), \quad \Pr(h \mid \text{ODD}) \approx f_e(h) p_h(h) (1 - p_a(h)), \quad (8)$$

where $L(h)$ is the loss or severity if the hazard occurs. The optional factorization uses exposure f_e , the conditional probability p_h of a hazardous event during exposure, and the probability p_a that the affected person can avoid harm. It is a screening model only when these terms are estimated as conditional probabilities. Multiplying ordinal risk ranks does not produce a calibrated probability and does not replace the risk assessment required by the applicable standard. Workspace layout changes exposure, sensing and control architecture change p_h , robot speed and geometry change loss severity, and warnings or escape space change avoidability. Model quality is therefore one design lever rather than the sole source of risk [16, 127]. Some hazards arise when the intended function operates in unforeseen conditions rather than when a component fails. This is an important failure mode for learned perception. Such hazards fall under the safety of the intended function framing and require scenario based rather than failure rate based analysis [128–131].

Latency is a safety parameter, not only a performance one. The most concrete illustration of why Physical AI design differs from traditional AI design is the speed and separation monitoring criterion for collaborative operation. It requires that the separation between the human and robot never fall below the protective distance [104, 132]

$$S_p(t_0) = S_h + S_r + S_s + C + Z_d + Z_r, \quad (9)$$

where S_h is the distance the human can travel while the system reacts and stops, S_r the robot's travel during its reaction time T_r , S_s its braking distance, C the intrusion distance of a body part, and Z_d, Z_r the position uncertainties of the human-detection system and the robot. Under constant human speed v_h , robot speed v_r , and deceleration $a_{\max}$, this reduces to

$$S_p = v_h(T_r + T_s) + v_r T_r + \frac{v_r^2}{2a_{\max}} + C + Z_d + Z_r, \quad T_s = \frac{v_r}{a_{\max}}, \quad (10)$$

and the system reaction time decomposes into

$$T_r = T_{\text{sense}} + T_{\text{perc}} + T_{\text{infer}} + T_{\text{comm}} + T_{\text{act}}. \quad (11)$$

The governance consequence is direct: model inference latency T_{infer} appears inside a safety related design distance. For illustration, let $v_h = 1.6 \text{ m/s}$, $v_r = 1.0 \text{ m/s}$, $a_{\max} = 2.0 \text{ m/s}^2$, $T_r = 0.2 \text{ s}$, and $C + Z_d + Z_r = 0.33 \text{ m}$. Equation 10 then gives $S_p = 1.90 \text{ m}$. Every additional 100 ms of reaction time widens the required separation by $(v_h + v_r)\Delta T = 0.26 \text{ m}$, which is about 14% of the original separation. These values illustrate the calculation only. An application must use measured stopping behavior, validated sensing uncertainty, the relevant intrusion allowance, and speeds selected through its risk assessment. Because Eq. 10 is quadratic in v_r , it can also be inverted to yield the maximum admissible robot speed for a given floor plan. Workspace area, actuator deceleration, sensor coverage, and model latency must therefore be traded jointly. This coupling is much less direct in most screen based AI services, where latency is usually a service quality concern rather than part of a physical safety envelope.

Contact limits bound commanded velocity. Where separation cannot be maintained, power and force limiting applies instead. For a transient contact between a human body region of effective stiffness k and mass m_H and a robot of effective mass m_R , the reduced mass and the resulting relation between force and velocity are

$$\mu = \left(\frac{1}{m_H} + \frac{1}{m_R} \right)^{-1}, \quad F = v_{\text{rel}} \sqrt{k \mu} \implies v_{\text{rel}}^{\max} = \frac{F_{\max}}{\sqrt{k \mu}}, \quad p = \frac{F}{A} \leq p_{\max}, \quad (12)$$

with $F_{\max}$ and $p_{\max}$ specified per body region and A the contact area determined by end-effector geometry [103, 104]. This idealized energy balance provides a preliminary velocity bound and shows why padding, compliant actuation, and rounded

end-effector geometry can reduce contact severity. It does not replace application specific contact testing, biomechanical limits, or the risk assessment required by the relevant robot safety standard.

Design the enforcement layer, not only the policy. Because a learned policy cannot generally be shown to avoid every unsafe proposal, the design phase should introduce an explicit enforcement layer between policy output and actuation. For control affine dynamics and a safe set $\mathcal{C} = \{x : h(x) \geq 0\}$ encoded by a barrier function h , the nominal policy action $u_\pi(x)$ can be projected onto the admissible set by solving, at every control step,

$$u^\star = \arg \min_{u \in \mathcal{U}} \|u - u_\pi(x)\|^2 \quad \text{s.t.} \quad L_f h(x) + L_g h(x) u \geq -\alpha(h(x)), \quad (13)$$

so that the filter, rather than the learned model, enforces the encoded safety property. A simplex architecture that switches to a verified fallback controller when the state approaches the boundary of the recoverable set serves a similar role at the architecture level [105, 133, 134]. A conformal score can provide a separate trigger for escalation. Given n exchangeable calibration examples with nonconformity scores $s_1, \dots, s_n$, the $[(n+1)(1-\alpha)]$ order statistic defines a threshold $\hat{q}$. Under exchangeability, the next score exceeds this threshold with probability at most approximately α [135]. This is a marginal coverage statement, not an automatic bound on unsafe behavior. A safety interpretation additionally requires a score that has been validated against the relevant hazard, and the guarantee may degrade under temporal dependence or distribution shift. The design review should therefore state which hazards are handled by the safety filter, which trigger human escalation, and which remain outside the validated envelope.

Co-design under joint budgets. Physical AI design requires the co-design of AI models, software, and hardware rather than optimizing each component independently. AI design determines the learning algorithms, perception, reasoning, planning, and control capabilities; software design provides the system architecture, middleware, communication, and real-time execution; while hardware design encompasses the mechanical structure, sensors, actuators, embedded computing, and power systems that enable embodied intelligence [28, 44, 136]. Three budgets couple these layers quantitatively. The control loop must sample well above the closed-loop bandwidth f_{bw} , conventionally $f_s \gtrsim 10 f_{\text{bw}}$. Many current large multimodal policies cannot run directly at that rate, which motivates a hierarchical split between a slower policy with frequency f_π , often around 5 to 10 Hz, and a faster low level controller, often around 10^2 to 10^3 Hz. Bridging that split with action chunking imposes a hard real-time feasibility condition on the chunk horizon H executed at rate f_{exec} ,

$$\frac{H}{f_{\text{exec}}} \geq T_{\text{infer}} \iff H \geq T_{\text{infer}} f_{\text{exec}}, \quad (14)$$

so that the next chunk is available before the current one is exhausted; violating Eq. 14 produces control gaps that manifest as jerk or stalls rather than as a degraded benchmark score [36]. Finally, the onboard energy budget over a task of duration T ,

$$E = \int_0^T (P_{\text{compute}}(t) + P_{\text{actuation}}(t)) dt, \quad E_{\text{decision}} \approx \frac{\overline{P}_{\text{compute}}}{f_\pi}, \quad (15)$$

couples policy frequency to endurance and thermal headroom, which is why dynamic-inference, pruning, and compression-recovery methods for embodied policies are design-stage rather than deployment-stage decisions: they change T_{infer} , and through Eq. 10 they change the safety envelope [137–139].

Human-centered and governance-by-design requirements. Human-centered design remains a fundamental principle throughout the design process. Rather than replacing humans during design, Physical AI systems should keep humans in the loop by incorporating stakeholder input, expert evaluation, and iterative human feedback to guide system development and improve safety, usability, and performance [140–142]. As companion and assistive robots become increasingly prevalent, design must extend beyond functional performance to include social interaction, user experience, trust, accessibility, and long-term acceptance. These considerations influence robots’ physical appearance, behavior, and interaction strategies, drawing increasingly on insights from psychology, interaction between humans and robots, and artificial intelligence to enable natural and effective collaboration with people [143–145]. In practice these commitments are discharged through named methods with reportable outputs, which Section 4.2 treats as a governance factor in its own right [24, 146]. Because such commitments are easily stated and rarely verified, we suggest treating governance requirements as first-class design requirements and tracking the fraction of them that carry an executable acceptance test,

$$C_{\text{gov}} = \frac{|\{g \in \mathcal{G} : \exists \text{ an executable test } t(g)\}|}{|\mathcal{G}|}, \quad (16)$$

where $\mathcal{G}$ is the set of governance requirements in the specification. A requirement such as “the robot shall yield to pedestrians” scores zero under Eq. 16 until it is restated as a measurable condition on S_p from Eq. 9; the metric thus makes the difference between a governance aspiration and a governance requirement observable at review time [95, 147].

A hypothetical worked design review. Consider a collaborative industrial manipulator that passes components to workers at a station where the maximum available separation along the approach direction is 1.5 m. The illustrative inputs above give

Table 5: Physical AI Data Taxonomy

Data Category	Definition	Capabilities Enabled	Applicable Models	Training Stage	Representative Research	References
Foundation & Human Knowledge Data	Internet-scale vision-language datasets, egocentric videos, language instructions, task annotations, and human demonstrations that provide semantic knowledge and behavioral priors for embodied intelligence.	Semantic grounding, visual representation learning, instruction following, affordance learning, commonsense reasoning, and long-horizon planning.	Vision-Language Models (VLMs), Vision-Language-Action (VLA) models, LLM planners, video foundation models.	Mainly pretraining.	PaLM-E, RT-1, OpenVLA, GR00T N1.	[72, 80–82]
Embodied Robot Experience	Robot observation-state-action trajectories collected through teleoperation, tactile sensing, proprioception, autonomous rollouts, and fleet experience.	Closed-loop control, dexterous manipulation, continual learning, morphology generalization, and policy adaptation.	Robot foundation models, diffusion policies, Vision-Language-Action models.	Pretraining, post-training, and continual learning.	RT-X, RoboCat, Open X-Embodiment.	[79, 80, 82, 148]
Synthetic & Simulation Data	Physics-engine simulations, digital twins, synthetic demonstrations, world-model rollouts, neural-generated trajectories, and simulated environments.	Counterfactual reasoning, scalable policy learning, sim-to-real transfer, rare-event generation, and planning.	World models, reinforcement learning, diffusion models, inverse dynamics models.	Pretraining, simulation, and post-training.	Genie, Isaac Sim, MuJoCo, ARIO.	[45, 66, 74, 98]
Spatial, Semantic & Reasoning Data	Scene graphs, spatial maps, semantic memory, task graphs, object relations, language grounding, and structured world knowledge.	Localization, navigation, spatial reasoning, instruction grounding, hierarchical planning, and task decomposition.	Spatial world models, multimodal VLAs, LLM planners.	Pretraining and post-training.	SayCan, PaLM-E, embodied world models.	[47, 48, 72, 76]
Evaluation, Safety & Reliability Data	Failure cases, intervention records, recovery trajectories, robustness benchmarks, uncertainty estimation, and safety annotations.	Capability evaluation, robustness assessment, uncertainty estimation, safety alignment, and deployment monitoring.	Safety critics, runtime monitors, evaluation frameworks.	Alignment and evaluation.	LIBERO-Safety, OpsieVerse, Safety-CHORES, and safety-instrumented LIBERO or RoboCasa evaluations.	[49, 149–151]
Governance & Ecosystem Data	Policies, regulations, standards, deployment records, patents, investment trends, supply-chain information, incident databases, and ecosystem indicators.	Life cycle governance, AI assurance, regulatory compliance, deployment analysis, and ecosystem intelligence.	Model-agnostic governance systems.	Usually outside model training.	UNESCO AI Recommendation, OECD AI Incident Monitor, CSET, Deloitte.	[1, 10, 152, 153]

a protective distance of 1.9 m, so this proposed layout cannot satisfy the separation requirement. The team can lower robot speed, improve braking, reposition the person detection sensors, restrict the shared workspace, or add a physical barrier. Each alternative changes a term in Eq. 10 and can be compared before hardware is ordered. If contact remains possible, Eq. 12 provides a preliminary bound for reviewing speed and end-effector geometry, followed by the application-specific testing required by the relevant standard. The review should also include operators and maintenance staff, since visibility, emergency-stop access, and the handover procedure affect whether the design is usable in practice. Each selected control becomes an acceptance test. The design leaves this stage only when the operational design domain is explicit, high severity hazards have an enforced control, mandatory governance requirements have full test coverage under Eq. 16, and the responsible owner has accepted the remaining risk [28, 104, 127].

Data: Data used in Physical AI can be organized into six overlapping categories: foundation and human knowledge data; embodied robot experience; synthetic and simulation data; spatial, semantic, and reasoning data; evaluation, safety, and reliability data; and governance and ecosystem data (Table 5). Different modalities enter the life cycle at different points and have distinct limitations. Web-scale video, for example, provides visual and semantic knowledge but usually lacks explicit action labels and measurements of physical interaction. A model trained mainly on passive video may therefore struggle to infer contact dynamics or the consequences of an action [46]. Combining complementary sources can improve both training efficiency and model performance. A data-pyramid strategy, for instance, can co-train on action-labeled robot trajectories and action-free observations [82, 148]. Continued collection across tasks, environments, and robot embodiments can then target gaps revealed during evaluation rather than simply increasing the total number of trajectories [47].

From data volume to risk-weighted coverage. A trajectory count alone is a poor governance measure because robot data are temporally correlated and common operating conditions can overwhelm rare but safety-critical cases. Before collection, a team can define a scenario set $\mathcal{C}$ across the dimensions that delimit its operational design domain. These dimensions may include task, environment, embodiment, user group, and hazard condition. The team then assigns each scenario cell c a risk weight w_c and a minimum evidence requirement $n_{\min,c}$. Dataset coverage can be summarized with a weighted score and a separate gate for safety-critical cells:

$$C_{\text{data}} = \frac{\sum_{c \in \mathcal{C}} w_c \mathbb{I}[n_c \geq n_{\min,c}]}{\sum_{c \in \mathcal{C}} w_c}, \quad G_{\text{crit}} = \min_{c \in \mathcal{C}_{\text{crit}}} \mathbb{I}[n_c \geq n_{\min,c}], \quad (17)$$

where n_c is the number of independent episodes in cell c and $\mathcal{C}_{\text{crit}}$ contains the safety-critical cells. The weighted score shows overall progress, while $G_{\text{crit}} = 1$ is a hard release condition. This separation prevents well-covered nominal cells from compensating for an uncovered high-risk cell. The weights should be fixed before evaluation from expected exposure and harm severity. The minimum $n_{\min,c}$ should be justified by a target confidence interval or rare-event bound, such as Eqs. 4 and 7, rather than chosen after observing the results. Repeated trajectories from one run should not be counted as independent evidence. Teams should use independently initialized sessions where possible and cluster-aware uncertainty estimates when episodes within a session remain correlated. Data quality is therefore assessed by the fitness for the intended use, not by the size of the dataset alone [154].

In a hypothetical hospital-robot example, one critical cell combines nighttime illumination, a wet floor, a crowded corridor, and a wheelchair user; 100,000 empty daytime runs do not cover it. A practical workflow is to (i) publish the scenario matrix, C_{data} , and G_{crit} ; (ii) split by site, period, and robot rather than adjacent frames; (iii) target uncovered critical cells while separating real and synthetic counts; and (iv) record configuration, calibration, consent, licenses, annotations, and interventions. Versioned lineage must link each policy to its exact training data and transformations [30, 155]. General task suites such as LIBERO, CALVIN, and RoboCasa are not safety evidence without collision, damage, intervention, and constraint-violation labels. LIBERO-Safety and OopsieVerse expose these outcomes but cannot replace application-specific testing [149, 150]. The dataset advances only when $G_{\text{crit}} = 1$, provenance and privacy checks pass, and no intolerable residual risk remains; otherwise Eq. 17 identifies the next collection target.

Model: The model stage converts curated data into learned components for perception, reasoning, planning, action generation, and control (Table 6) [2, 44]. The categories in the table describe complementary, overlapping roles and interfaces rather than a fixed sequence of modules. Vision-language foundation models provide semantic representations that align visual observations with language, while visual and spatial representation models emphasize object, geometric and spatial structure. Embodied reasoning models provide task decomposition, skill selection, and feedback-based replanning.

Robot foundation models are distinguished primarily by their broad training and transfer across robotic tasks, environments and embodiments, whereas vision-language-action (VLA) models are defined by their interface: they map visual observations, language instructions and optionally robot state to actions or action-oriented representations. Their outputs can include waypoints or intermediate spatial and control representations for downstream execution. Across model families, action generation combines output representations, such as continuous trajectories, discrete action tokens (including frequency-domain encodings), and latent actions, with generation mechanisms such as autoregressive decoding, diffusion, or flow matching [156–159]. Actions may be generated in chunks or organized hierarchically into subtasks and lower-level

commands [160, 161]. For example, the same model may be both a robot foundation model and a vision–language–action model, while using flow matching to generate action chunks. In humanoids and mobile manipulation systems, learned low-level or whole-body controllers can operate beneath VLA or reasoning models, translating motion targets and sensory feedback into motor commands for locomotion, balance, contact handling, and coordinated manipulation. Examples include Helix 02’s System 0, which tracks whole-body targets, and WholeBodyVLA’s loco-manipulation-oriented reinforcement learning policy, which executes locomotion commands [162, 163].

Within world and dynamics models, action-free predictive representations capture temporal structure without conditioning on actions and can support downstream control, while action-conditioned models explicitly predict future states or observations under candidate actions for planning and policy learning [164]. For example, V-JEPA 2 learns action-free predictive representations from video, whereas V-JEPA 2-AC adds action-conditioned post-training on robot trajectories to predict latent outcomes of candidate actions for robotic planning [165]. Generative world models synthesize visual environments or possible futures for simulation and synthetic experience generation, and may also be action-conditioned [166, 167]. Newer world–action models, such as DreamZero, jointly predict future observations and executable actions, linking world modeling with action generation [168].

Physical-system modeling and simulation models approximate physical processes for prediction, simulation, and system identification, using physics-informed neural networks, neural operators, or hybrid analytical–learned dynamics [169–172]. Physical knowledge can enter through equation-based training constraints, structured mechanics priors, or analytical dynamics augmented by learned residuals, while differentiable simulators enable gradient-based optimization of model parameters and policies [169, 172–174]. This category overlaps with world and dynamics models but also covers scientific and engineering simulation beyond embodied planning; neural operators need not explicitly enforce physical laws.

Architecture and model development establish the network structure, interfaces, learning objective, and action representation. Pretraining can build general representations from broad data, while mid-training specializes those representations to a target task, environment, or embodiment [81, 175, 176]. Post-training can further adapt behavior through task-specific fine-tuning, preference optimization, or safety-focused training [175]. Prototyping and sim-to-real transfer expose discrepancies between simulated and physical execution [79, 98]. Optimization then addresses latency, memory, energy, and hardware constraints, and testing determines whether the complete system meets the release criteria [66, 177]. These activities are iterative rather than a universal linear sequence. A failed physical test may send the team back to data collection, architecture changes, or safety-controller design.

Release gates for the complete system. Average task success is insufficient for selecting a Physical AI model because a gain on nominal tasks can hide unsafe actions in a rare scenario or a control-loop deadline miss. Using the safety-critical cells $\mathcal{C}_{\text{crit}} \subseteq \mathcal{C}$ defined during data curation, a candidate model should be admitted only if

$$\min_{c \in \mathcal{C}_{\text{crit}}} \text{LCB}_{1-\alpha}[p_{\text{succ}}(c)] \geq \tau_{\text{succ}}, \quad \max_{c \in \mathcal{C}_{\text{crit}}} \text{UCB}_{1-\alpha}[p_{\text{unsafe}}(c)] \leq \tau_{\text{unsafe}}, \quad \max_{c \in \mathcal{C}_{\text{crit}}} Q_{0.99}[T_{\text{loop}}(c)] \leq T_{\text{deadline}} \quad (18)$$

where T_{loop} includes sensing, inference, communication, and actuation, and all thresholds are fixed from the application risk analysis before model comparison. Equation 4 can be used for binary outcomes when trials are sufficiently independent. Correlated robot trials require session-level or cluster-aware bounds, as discussed in the Research component. The 99th-percentile latency estimate also needs enough observations in each critical cell and should be accompanied by uncertainty when the sample is small. All three conditions must pass: performance in an easy cell cannot compensate for a safety or timing violation in a critical one. Evaluation should additionally record human interventions, recovery success, energy per episode, and the sim-to-real gap. Task success and safe execution should be reported separately, using safety-specific benchmarks where applicable, but these diagnostic measures do not replace the release conditions [18, 49, 149, 150, 177].

For a hypothetical warehouse-picking decision, suppose that each model is evaluated in 1,000 sufficiently independent human-present trials. Model A reaches 94% average task success, produces 12 unsafe actions, and has 180 ms 99th-percentile loop latency. Model B reaches 91% average success, produces 2 unsafe actions, and has 60 ms latency. The observed unsafe-action rates are 1.2% and 0.2%, but the release gate uses uncertainty rather than these point estimates. A one-sided 95% Wilson calculation gives upper bounds of approximately 1.91% for A and 0.60% for B. With $\tau_{\text{unsafe}} = 1\%$ and a 100 ms deadline, A fails both the safety and timing gates. B passes those two gates but remains ineligible until its minimum per-cell success bound also passes. Teams can implement the gate in three steps. First, evaluate the incumbent and candidate on the same versioned scenario matrix. Second, stress-test sensor dropout, delayed observations, ambiguous instructions, unseen objects, and actuator saturation. Third, attach the per-cell results, model checkpoint, training-data version, hardware target, and threshold decisions to a release record. Model validation then becomes an auditable deployment decision rather than a leaderboard comparison.

Table 6: Physical AI Model Taxonomy.

Category	Representative Models	Primary Role	References
Vision–Language Foundation Models	CLIP, SigLIP, LLaVA, PaLI-X, PaliGemma 2, Qwen3-VL	Provide aligned visual–language representations and semantic knowledge to support embodied perception, instruction interpretation, and reasoning.	[72, 178–182]
Visual & Spatial Representation Models	DINOv3, SAM 3.1, Grounding DINO, VGGT, Depth Anything 3, R3M, MVP	Extract visual features, object information, and scene geometry to support recognition, segmentation, grounding, localization, and manipulation.	[43, 44, 73, 183–187]
Robot Foundation Policies	Octo, RoboCat, RT-1, RT-X, RT-H, GR00T N1, GR00T 1.7, π_0 , $\pi_{0.5}$, Gemini Robotics, MolmoAct 2	Learn reusable manipulation and locomotion skills from broad, multi-embodiment robot datasets to support cross-task generalization and adaptation across tasks, environments, and embodiments.	[44, 79, 80, 82, 148, 159, 161, 188–191]
Vision–Language–Action Models	RT-2, OpenVLA, OpenVLA-OFT, π_0 , $\pi_{0.5}$, CogACT, GR00T N1, GR00T 1.7, MolmoAct 2, Gemini Robotics, Gemini Robotics 2, Helix	Map visual observations, language instructions, and optional robot states to actions or intermediate action representations for execution through downstream decoders or controllers.	[81, 82, 157, 159, 161–163, 189–196]
World & Dynamics Models	DreamerV3, V-JEPA 2, V-JEPA 2-AC, Genie 3, Cosmos-Predict2, Cosmos Predict, DreamZero, Video World Models, Robot Dynamics Models	Learn action-free predictive representations, predict action-conditioned dynamics, generate possible environments and futures, or jointly predict observations and actions.	[2, 45, 74, 164–168, 197]
Learning/Adaptation Method	$\pi_{0.6}^*$ (RL post-training); RoboTTT (test-time training); Skild S1, Generalist GEN-1.5 (in-context/physical prompting)	Adapt policy behavior after pretraining through reward-based reinforcement learning, partial online updates at inference time, or in-context demonstrations, to support post-deployment correction and rapid task acquisition without full retraining.	[198–201]
Deployment & Efficiency	Shallow- π ; SpecPrune-VLA, RLRC; DeeR-VLA (dynamic depth/early exiting); LightDP; SmolVLA; SmolVLA	Reduces a model’s latency, memory, and compute cost for real-time control on resource-limited robotic hardware, via pruning, quantization, dynamic-depth inference, asynchronous pipelines, or on-device deployment.	[138, 195, 202–205]
Physical-System Modeling & Simulation Models	PINNs, Fourier Neural Operator, DeepONet, Hamiltonian Neural Networks, hybrid residual dynamics models	Approximate physical processes for prediction, simulation, and system identification using learned surrogates, physical constraints, or combined analytical and learned dynamics.	[169–174]
Embodied Reasoning & Planning Models	SayCan, Inner Monologue, PaLM-E, RoboBrain 2.0, Gemini Robotics ER 2, MolmoAct 2, Cosmos Reason 2, ThinkAct, STEER, Hume	Support goal interpretation, spatial reasoning, task decomposition, skill selection, and planning or replanning from execution feedback.	[72, 76, 78, 190, 191, 206–209]
Physics-Informed & Hybrid Dynamics Models	PINNs, Hamiltonian Neural Networks, Neural ODEs, Fourier Neural Operator (FNO), DeepONet, DiffTaichi, Brax, MuJoCo MJX, AC-HGN	Incorporate physical laws, governing equations, conservation principles, differentiable physics, or learned dynamics into model learning.	[173, 210–214]
Low-Level Physical Control Models	ANYmal Neural Control Policy, DeepMimic, Neural Motor Policies, Neural Inverse-Dynamics Models, Physics-Based Control Policies	Translate physical states, desired motion, or high-level commands into executable motor actions, joint trajectories, forces, or torques while accounting for system dynamics, feedback, and real-time physical constraints.	[215, 216]

Deployment: The deployment stage integrates a validated model into a complete physical system and manages that system throughout operation. System integration and end-to-end testing connect the model to sensing, communication, safety logic, and actuation [49, 136]. Hardware commissioning calibrates sensors and actuators, while system identification estimates properties such as dynamics, friction, and mass distribution for the actual robot [66]. Precommissioning simulation and controller tests can expose integration failures before physical trials [45, 80]. Policy evaluation then repeats the safety-critical scenarios on the target hardware and in the intended operating environment. After release, operational monitoring, maintenance, and controlled updates are required because wear, calibration drift, firmware changes, and environmental changes can invalidate earlier evidence [21, 217]. A change to any safety-relevant component creates a new deployable unit that must be assessed against the approved configuration and operating domain.

Deployment as a monitored safety case. Commissioning should freeze and identify the complete deployable unit. This unit includes the model checkpoint, preprocessing and post-processing code, runtime container, robot and sensor firmware, calibration, safety controller, and operational design domain. The team should then establish baseline measurements on the same scenario matrix used at the data and model stages. Runtime monitoring must combine hard physical invariants with leading indicators of degradation. One compact alarm score over a rolling window W is

$$A_t = \max \left\{ \frac{[S_p(t) - d(t)]_+}{S_p(t)}, \frac{[Q_{0.99}(T_{\text{loop}}; W) - T_{\text{deadline}}]_+}{T_{\text{deadline}}}, \frac{[\text{UCB}_{1-\alpha}(r_{\text{int}}; W) - \tau_{\text{int}}]_+}{\tau_{\text{int}}}, \frac{[e_{\text{cal}}(t) - \tau_{\text{cal}}]_+}{\tau_{\text{cal}}} \right\}, \quad (19)$$

where $[x]_+ = \max(x, 0)$, $d(t)$ is the measured separation between the human and robot, $S_p(t)$ is the protective distance from Eq. 9, r_{int} is the rate of human interventions or emergency stops, and e_{cal} is calibration error. The denominators and the confidence level for the intervention-rate bound must be positive and fixed during commissioning. The window W must also contain enough independent operational units to support the chosen quantile and confidence bound. The score A_t is a dimensionless alarm index, not an estimate of accident probability or proof of safety. A value of zero only means that these four monitored signals have not crossed their thresholds; unmonitored hazards may still be present. A separation violation should immediately invoke the verified safety response or stop the system. A timing violation should prevent the learned controller from issuing further actuation commands unless an independent safety channel continues to enforce the physical invariant. Rising intervention or calibration signals can first trigger a degraded mode, such as reduced speed, a restricted workspace, or mandatory human confirmation [35, 95, 105].

For a hypothetical deployment example, suppose a camera firmware update on a mobile manipulator leaves task success at 96% but raises the 99th-percentile loop latency from 72 ms to 128 ms against a 100 ms deadline. The timing term in Eq. 19 becomes 0.28, so the system should not wait for a collision or a fall in average success. It should restrict motion or enter the safety controller, freeze the rollout, capture the sensor and timing logs, and restore the last validated bundle of firmware, model, and calibration settings. Model and software updates should follow four steps. First, run a shadow evaluation without actuation. Second, make a canary release to a small fleet or restricted operating domain. Third, compare the update with the incumbent on success, unsafe actions, latency, interventions, and energy use in every scenario cell. Fourth, promote or automatically roll back the update according to predeclared thresholds. Every transition should leave a tamper-evident, versioned record linking the event to the deployed components and the evidence that authorized the change [30, 95, 96]. Incidents should be added to the scenario matrix and routed back to the Data and Model stages instead of being handled as undocumented maintenance patches.

4.2 Operationalizing Governance Elements into Each Life Cycle Component

Operationalization means converting a governance value into a testable control at a named life cycle decision. Prior life cycle-oriented trustworthy-AI work, the NIST AI Risk Management Framework, and ISO/IEC 42001 provide process-level scaffolding for governing, mapping, measuring, and managing risk, but they do not by themselves specify the physical evidence needed for a particular robot, operating environment, or human-robot interaction [15, 218, 219]. We therefore represent the operationalization of each governance element g by a control contract

$$\Gamma_g = (\text{owner}_g, \text{artifact}_g, m_g, \tau_g, \text{evidence}_g, \text{trigger}_g), \quad (20)$$

where the owner has decision authority, the artifact is a versioned deliverable, m_g is a measurable quantity, τ_g is a predeclared acceptance threshold, the evidence records the test result, and the trigger specifies when the evidence expires. Typical triggers for Physical AI include a change to the model, middleware, sensor, firmware, calibration, end effector, human-oversight procedure, or operational design domain (ODD), as well as an incident or statistically significant drift in field performance. At every life cycle handoff, the owner should record a go, conditional-go, or no-go decision and link it to the exact system configuration and evidence. This makes governance continuous without reducing it to continuous paperwork: review is repeated when risk-relevant facts change, not on an arbitrary schedule alone [95, 218].

Table 7 organizes the resulting elements into three tiers: Fundamental factors that apply across all stages; Knowledge Generation (Research and Design); and Building, Deployment, and Operation (Data, Model, and Deployment). The

Category		Governance Element	Representative References
Fundamental		Reliability	[99, 122, 218]
		Safety Assurance	[18, 104, 127, 220]
		Human Value Alignment	[28, 141, 221]
		Transparency and Accountability	[218, 222–224]
Knowledge Generation	Research	Open Science	[101, 225]
		Reproducibility	[100]
	Design	Inclusiveness	[226, 227]
		Generalizability	[100, 121]
Building, Deployment, and Operation	Data	Privacy and Transparency	[25, 228]
		Traceability	[30, 222]
	Model	Explainability	[229, 230]
		Bias and Fairness	[231, 232]
	Deployment	Resilience	[96, 233]
		Sustainability	[38, 234]

Table 7: Governance elements organized by category, sub-category, and supporting citations across the Physical AI life cycle.

subsections below instantiate Eq. 20 with artifacts, metrics, benchmarks, and revalidation triggers. Benchmarks are treated as diagnostic evidence rather than universal certificates: results from a simulator, dataset, or reference robot must still be revalidated for the target embodiment and ODD.

4.2.1 Fundamental Trustworthiness Factors

The four fundamental factors apply at every life cycle stage and cannot be delegated to a final model audit. They are treated as fundamental because failure of any one can invalidate evidence or responsibility at every subsequent handoff, whereas the remaining elements in Table 7 have a primary life cycle owner. They answer four review questions: Does the evidence remain valid under realistic variation (Reliability)? Can identified hazards be prevented, detected, and brought to a safe state (Safety Assurance)? Do system requirements and evaluations reflect the people who use, supervise, and are affected by the system (Human Value Alignment)? Can a safety-relevant action be reconstructed and assigned to a responsible decision-maker (Transparency and Accountability)? Following Eq. 20, each factor should be operationalized through a versioned artifact, a predeclared acceptance criterion, recorded evidence, and an explicit revalidation trigger.

Reliability asks whether the *complete configured system* continues to perform its required function for a stated duration and under the conditions in its ODD; it is not equivalent to a high average benchmark score. Embodiment makes this distinction operationally important because lighting, friction, payload, calibration, sensor contamination, network delay, and actuator wear can change the same policy’s behavior without changing its weights [102, 235]. RobotArena ∞ supports controlled perturbation of manipulation scenes, while S2R-Bench measures autonomous-driving perception under real sensor anomalies, weather, lighting, and road variation [99, 122]. These resources are useful templates for stress testing, but neither certifies a different robot or ODD.

A practical reliability workflow has three steps. First, define a versioned scenario matrix over task, environment, embodiment, human presence, and degradation conditions, and mark the safety-critical cells before testing. Second, run independently initialized trials in simulation and on the target hardware, reporting run-level failures, the Wilson intervals in Eq. 4, the stage-transfer gap in Eq. 3, and tail rather than mean control-loop latency. Third, gate the system on the weakest critical cell using Eq. 18; a strong nominal cell must not compensate for a failed critical one.

Writing $p_{\text{succ}}(c \mid v)$ for the per-episode success probability in cell c of the scenario matrix ($\mathcal{C}$ in Eq. 17) under the deployed configuration v (model, software, firmware, calibration, and hardware; the bundle later logged as v_t in Eq. 23), the aggregate

score and the reliability gate are

$$\bar{p}(v) = \sum_{c \in \mathcal{C}} \omega_c p_{\text{succ}}(c | v), \quad G_{\text{rel}}(v) = \mathbb{I} \left[\min_{c \in \mathcal{C}_{\text{crit}}} \text{LCB}_{1-\alpha}[p_{\text{succ}}(c | v)] \geq \tau_{\text{succ}} \right], \quad (21)$$

where ω_c is the fraction of episodes that fall in cell c (exposure, not the risk weight w_c of Eq. 17) and the right-hand condition is the first condition of Eq. 18. The score is a weighted sum, so a rarely visited cell can fail badly without moving it; the gate is a minimum, so no cell can be compensated.

For example, a warehouse policy with 96% aggregate success but 78% success under simultaneous low light and partial shelf occlusion (a cell with $\omega_c = 0.05$ and 97% success elsewhere gives exactly this $\bar{p}$) should be held at the current stage if the predeclared threshold τ_{succ} on that cell’s lower confidence bound is 90%. The team can then collect targeted data, revise sensing or illumination, or restrict the ODD. A passed result remains valid only for the recorded model, firmware, calibration, end effector, and ODD; changing any of these triggers revalidation [102, 218, 235].

Safety Assurance starts from hazards and system boundaries, not from a generic safety benchmark. A complete safety-assurance process should specify the ODD and reasonably foreseeable misuse, identify hazards, estimate and evaluate risk, apply the risk-reduction hierarchy, and record the residual risk and its accepting authority. ISO 12100 supplies this machinery-level workflow; application-specific standards then constrain the implementation, such as ISO 10218 for industrial robots, ISO/TS 15066 for collaborative operation, and ISO 21448 for hazards caused by performance limitations of an intended function in road vehicles [104, 127, 128, 220]. This ordering prevents a perception score or language-model refusal rate from being mistaken for evidence that a complete Physical AI system is safe.

Each intolerable hazard should become a row in a living hazard log containing its causal scenario, affected person or asset, severity and exposure assumptions, preventive control, runtime detector, fallback or safe state, verification test, evidence link, owner, and revalidation trigger. The argument connecting these claims to evidence can be organized as a Goal Structuring Notation safety case. Learned policies should be separated from independently enforced constraints wherever practicable: the safety filter in Eq. 13 can reject an unsafe proposal, and a simplex architecture can transfer control to a verified fallback before the recoverable set is left [105, 133]. To keep this safety argument hazard-indexed rather than model-score-indexed, the extent of that separation can be summarized over the hazard log itself. Writing $\mathcal{H}_{\text{crit}} \subseteq \mathcal{H}$ for the hazards judged intolerable in the risk assessment,

$$C_{\text{enf}} = \frac{1}{|\mathcal{H}_{\text{crit}}|} \sum_{h \in \mathcal{H}_{\text{crit}}} \mathbb{I}[h \text{ has a policy-independent safety barrier}], \quad (22)$$

where a policy-independent barrier is one whose actuation does not rely on the nominal learned policy, such as a safety-rated stop, a mechanical limit or interlock, an independently enforced speed or force restriction, or an independently evaluated safety filter. This quantity is not a residual-risk metric or a certification threshold, and a barrier counted here is only as dependable as the sensing coverage, timing, and integrity of the channel that carries it. Equation 22 makes explicit which critical hazards still rely on the learned policy as part of their safety argument. For these hazards, a change in π_t can invalidate part of the evidence supporting the current safety claim and may therefore require renewed evidence, whereas hazards protected by independent enforcement are primarily revalidated when the enforcing channel, its sensing coverage, timing assumptions, or the ODD changes [16, 133, 220]. Conformal monitors can provide calibrated warning-rate guarantees under their stated exchangeability assumptions, but such guarantees apply to the monitor’s score and do not by themselves prove system safety under distribution shift [135].

Benchmarks should be chosen by hazard. LIBERO-Safety separates task success from physical and semantic violations in VLA manipulation, while OopsieVerse measures mechanical, thermal, and fluid damage during household manipulation [149, 150]. They are valuable for finding failure modes, but simulation results must be followed by target-hardware tests of stopping time, separation, contact force or pressure, collision energy, and fallback activation as applicable. For example, the hazard “unexpected human entry during a manipulator handover” should be linked to a human-detection requirement, the protective-distance calculation in Eq. 9, a safety-rated stop, measured worst-case reaction and stopping times, and a named owner. A camera, firmware, layout, or policy update could invalidate the linked evidence and reopen that hazard. The release decision is therefore based on a traceable safety case for the complete configuration, not on an assertion that the learned model is safe [18, 132].

Human Value Alignment Physical AI turns values such as autonomy, accessibility, privacy, dignity, appropriate trust, and meaningful human control into properties of a shared physical interaction[221]. Operationalization of human values into physical AI therefore must consider all people affected by a physical system: primary users, operators, maintainers, caregivers, bystanders, and people exposed without choosing to interact [236]. Value Sensitive Design, participatory design, and human-centered design processes provide methods for eliciting these auxiliary users’ interests and resolving conflicts early [141]. Co-design studies with people with mobility disabilities show why accessibility requirements cannot be inferred

Table 8: Behavioral Measures

Behavioral Measure	Definition	Citation
Missed or unwanted interventions	An intervention is triggered when the autonomy’s inferred intended action diverges from the user’s measured command under sufficiently low predictive uncertainty: $\phi_{\text{inferred}}^t \neq \phi_m^t$ and $H(p(a^t \phi_m^t)) < \epsilon$, where $p(a^t \phi_m^t) = \eta p(a^t) \sum_{\phi_i^t \in \Phi} p(\phi_m^t \phi_i^t) p(\phi_i^t a^t)$. An intervention is unwanted when it is triggered despite the measured command reflecting the user’s true intent.	[238]
Legibility (fraction of tasks in which intent was inferable and correctable)	The time-weighted probability that an observer correctly infers the robot’s intended goal G^* from a partial trajectory: $\text{legibility}(\xi) = \frac{\int P(G^* \xi_{S \rightarrow \xi(t)}) f(t) dt}{\int f(t) dt}$, where $f(t)$ weights earlier segments of the trajectory more heavily.	[239]
Mode-confusion errors	An error that occurs when a user’s belief regarding the currently active control mode diverges from the robot’s true active mode, resulting in a command appropriate to an unintended mode.	[240]
Task and recovery success by user group	The proportion of task attempts completed successfully, and the corresponding proportion of failure trials from which the system or user recovers, computed separately per user subgroup (e.g., by impairment level).	[241]
Override success	The proportion of user override or correction attempts that successfully redirect robot behavior toward the user’s intended goal.	[242]
Time to reach a safe state after an override	The elapsed time between an override or stop command and the moment at which the robot reaches a verified safe state (e.g., zero velocity or a retracted, safe configuration).	[104]

solely by developers [28, 237]. The output of co-design process should be a versioned value-to-requirement matrix in which each material value has a stakeholder, system requirement, foreseeable failure, acceptance test, responsible owner, and method for contesting or revising the requirement. Tools such as the HRI Value Compass can support elicitation, but the resulting values become governable only after they are linked to testable requirements [221].

Evaluation should combine behavioral and self-reported evidence. Behavioral measures can include task and recovery success by user group [241], missed or unwanted interventions [238], override success [242], time to reach a safe state after an override [104], mode-confusion errors [240], and the fraction of tasks for which a person could understand and change the robot’s intended action [239]. Subjective instruments should be selected for the construct rather than treating “trust” as a universal score: NASA-TLX measures perceived workload, while the Trust Perception Scale-HRI is designed for perceived trust in human–robot interaction [243, 244]. Results should be reported with uncertainty and disaggregated across relevant abilities, experience levels, and interaction contexts. High trust is not itself the target; calibrated reliance is, so a participant who refuses an unsafe robot recommendation can represent a successful outcome.

For example, an assisted-living service robot should be reviewed with residents, caregivers, visiting family, maintenance staff, and bystanders [245]. A value such as autonomy can be translated into notice before recording, reversible task choices, an accessible stop or decline mechanism, and escalation to a caregiver only under predeclared conditions [246]. Before a pilot, the team sets thresholds for stop-command detection, time to safe state, task completion across mobility groups, and operator workload; records both failures and workarounds; and names the person who can accept a residual value conflict [247]. A change to the user population, autonomy mode, interface, or escalation policy triggers a new Human-Robot Interaction (HRI) evaluation [248]. An HRI evaluation refers to a structured and user-centered evaluation of how a change affects the interaction between humans and the robot, including relevant dimensions such as usability, social acceptance, user experience, workload, trust, and reliance [249]. It employs quantitative methods, including task-performance metrics, behavioral measures, standardized scales, and statistical analysis; qualitative methods, including interviews, observations, and open-ended responses; or mixed-methods approaches that integrate quantitative and qualitative evidence to assess complementary dimensions of human–robot interaction [250]. This converts “human-centered” from a design aspiration into evidence that can block or condition a release [218, 251].

Transparency and Accountability solve different but coupled problems. Transparency determines what information each stakeholder can obtain before, during, and after operation; accountability determines who has authority and responsibility

to act on that information. IEEE 7001 accordingly treats transparency as stakeholder-specific and measurable, while the NIST AI RMF requires documented roles, escalation paths, go/no-go decisions, and executive responsibility for risk decisions [218, 222]. For Physical AI, a model card alone is insufficient because a physical action depends on the sensor observation, estimated world state, nominal policy output, safety-layer modification, control mode, and deployed hardware and software configuration.

For every safety-relevant event at time t , a minimal event record can be represented as

$$\ell_t = (t, o_t, \hat{x}_t, u_t^\pi, u_t^{\text{safe}}, m_t, v_t), \quad (23)$$

where o_t identifies the sensor observation or an integrity-protected reference to it, $\hat{x}_t$ is the estimated state, u_t^π is the learned policy’s proposed action, u_t^{safe} is the action after safety filtering, m_t is the autonomy or fallback mode, and v_t identifies the deployed model, software, firmware, calibration, and hardware bundle. Records should use synchronized clocks, integrity protection, access controls, and retention periods appropriate to the hazard and privacy risk. The EU AI Act requires automatic event logging for covered high-risk AI systems, and UN Regulation No. 157 provides a domain-specific example by requiring timestamped records that clarify control transitions, system activation, and overrides in automated lane-keeping systems [223, 224]. Data sovereignty and minimization remain important, but belong in the privacy control: transparency does not require retaining every raw sensor stream indefinitely.

We propose *trace completeness* as a simple audit gate. If $\mathcal{E}_{\text{crit}}$ is the set of sampled safety-critical events and $\mathcal{F}_e$ the fields required to reconstruct event e , then

$$C_{\text{trace}} = \frac{\sum_{e \in \mathcal{E}_{\text{crit}}} \sum_{f \in \mathcal{F}_e} \mathbb{I}[f \text{ is present and integrity-verified}]}{\sum_{e \in \mathcal{E}_{\text{crit}}} |\mathcal{F}_e|}. \quad (24)$$

Before release, teams should require $C_{\text{trace}} = 1$ for the sampled critical events and complete an incident-replay drill within a predeclared time budget. This metric measures record completeness, not explanation quality or causal certainty.

Accountability is then attached with a responsibility matrix spanning the model provider, hardware manufacturer, system integrator, deployer, and site operator: every life cycle gate and incident class needs at least one responsible party and exactly one accountable decision owner. For example, after an unexpected manipulator motion, the team should be able to replay the chain from observation through safety-filtered command to actuator, identify whether the failure arose in calibration, inference, integration, or operating procedure, and route the corrective action to its owner. A missing record, ambiguous owner, or failed replay is a no-go result, even if the robot’s average task score remains unchanged [30, 109].

4.2.2 Research Trustworthiness Factors

Research governance establishes the scientific foundation upon which trustworthy Physical AI systems are built. Unreliable or irreproducible research can spread into downstream engineering decisions and eventually affect real-world safety because Physical AI systems operate in highly dynamic and safety-critical environments.

Open Science covers the practice of making research artifacts, including code, datasets, hardware designs, and experimental protocols—openly accessible to enable independent verification, reuse, and extension by other researchers [119]. In traditional AI governance, open science is largely confined to the release of source code, trained model weights, and digital datasets, since the artifacts under study are themselves purely digital and can be shared at near-zero marginal cost. Physical AI governance extends open science to the additional burden of hardware transparency: reproducing a result in embodied AI often requires not only open code but open mechanical designs, bills of materials, sensor specifications, and assembly documentation [252, 253], since the physical platform itself is frequently a confound that closed-source or proprietary hardware makes impossible for other labs to control for [100]. Open hardware in robotics therefore carries a documentation burden with no digital analogue, and community efforts to standardize how mechanical designs, part sourcing, and assembly procedures are published are what make a released design usable by a second lab rather than merely visible to it [225, 254]. The goal of this governance factor is to lower the barrier to independent replication and cross-lab comparison in Physical AI research, extending open science norms from the digital artifact to the physical system that produced it.

Openness in Physical AI, however, is not simply the digital release debate with hardware attached. Traditional discussions of staged release and structured access concern the diffusion of *informational* capability, and the harms they anticipate are mediated by a human who acts on the model’s output. What an embodied release confers is closer to operational capability: a mechanical design, a control stack, and a policy checkpoint together constitute a system that acts on the world directly, and the robotics software ecosystem in which such releases circulate has documented exploitable weaknesses of its own [9, 20, 21]. The concern is not hypothetical, since language-conditioned robot policies have been shown to be steerable into unsafe physical actions that their language backbone would refuse to describe [255]. The resolution is not to release less but to be precise about what each artifact is for. Artifacts that make a claim checkable—mechanical designs, calibration and evaluation protocols, task definitions, benchmark scripts—are largely distinct from artifacts that make a

system deployable, such as tuned safety envelopes, site-specific configurations, and operational fleet data; publishing the former is what open science requires, while the latter can be made available to independent auditors under structured access [33] without open publication. Framing the question this way lets a research community satisfy replication and proliferation concerns simultaneously, rather than treating them as a trade-off in which one norm must give way.

Reproducibility, closely related to reliability, ensures that independent researchers can replicate published findings using the same methodology. The robotics community has documented its own version of the reproducibility crisis: closed-source, simplified, or specially tailored hardware and software in laboratory demonstrators renders reproduction of most published robotics experiments difficult, a concern serious enough that a unified conceptual framework (MIRRER) has been proposed specifically to formalize and jointly evaluate reproducibility, replicability, and generalizability in robotics research [100]. The problem is not confined to any one subfield—reviews of surgical robotics have likewise found that reproducibility is not yet mainstream practice, with patenting restrictions, safety constraints, and closed hardware compounding the difficulty of replication [256], while IEEE Robotics & Automation Magazine’s Reproducible Articles initiative was created specifically to encourage detailed, replicable experimental protocols in intelligent robotics [120]. For Physical AI, reproducibility extends well beyond releasing source code. A minimal disclosure set for an embodied result comprises: (i) the hardware bill of materials, including robot model, end effector, and sensor part numbers and firmware versions; (ii) the calibration procedure and its residual error; (iii) the controller type and control frequency; (iv) the simulator version and its physics parameters; (v) datasets, preprocessing pipelines, hyperparameters, random seeds, and released checkpoints; and (vi) the number of trials, the success criterion, and the interval from Eq. 4. Items (i)–(iii) have no analogue in traditional AI reproducibility checklists, and they are precisely the ones that closed or specially tailored hardware makes impossible to satisfy [100, 252].

Disclosure alone, however, does not say when a replication has succeeded. If the originating laboratory reports a success rate $\hat{p}_A$ over n_A trials and an independent group reports $\hat{p}_B$ over n_B trials, the reproducibility gap

$$\Delta_{\text{repro}} = |\hat{p}_A - \hat{p}_B| \quad (25)$$

is itself a reportable statistic, and a replication should be judged successful when the Wilson intervals of the two estimates overlap rather than when a demonstration video looks similar. Consider a concrete case: a laboratory publishes a cloth-folding policy fine-tuned from an open-source vision–language–action checkpoint, reporting an 82% success rate on a 7-DoF Franka arm with a parallel-jaw gripper, and a second group reproduces it at 61% on a UR5e with a different gripper and camera mounting. Without the disclosure set, that $\Delta_{\text{repro}} = 21$ points is uninterpretable: it is consistent with a fragile method, with an 8 mm hand–eye calibration residual, with a control loop running at 20 Hz instead of 50 Hz, or with a gripper whose finger compliance differs enough to change the contact dynamics of the fold. With items (i)–(iv) disclosed, each of these can be checked and ruled in or out; without them, the community can only record that the result “did not reproduce” and lose the information about why. The second group should therefore report Δ_{repro} together with its own trial count and platform deltas rather than a bare success rate, so that a discrepancy is attributable to the method rather than to an undocumented difference in platform. Such practices improve scientific credibility while accelerating cumulative progress across the Physical AI research community.

4.2.3 Design Trustworthiness Factors

Governance should be embedded during the system design phase before implementation decisions constrain future development. Early design decisions determine system objectives, operational boundaries, acceptable risks, and stakeholder priorities, thereby influencing downstream data collection, model development, and deployment strategies.

Inclusiveness seeks to ensure that Physical AI systems reflect the needs, expectations, and values of the diverse individuals and communities that interact with them rather than optimizing exclusively for technical performance. Value-sensitive approaches applied directly to embodied systems illustrate how this can be done in practice: participatory design studies for socially assistive robots have used structured value-elicitation methods with real stakeholder groups to surface conflicting priorities before a system is built, rather than after deployment [226], and multi-stakeholder empirical studies of socially assistive robots in elder care have shown that patients, family caregivers, and clinical staff often prioritize different values (autonomy versus safety, companionship versus efficiency), values that only surface through direct, structured elicitation [227]. Operationally, inclusiveness can be incorporated through stakeholder mapping, participatory design workshops, interviews, usability testing, Wizard-of-Oz prototyping to elicit interaction expectations before a policy exists, accessibility review against published interaction guidelines, and iterative human-in-the-loop design reviews involving both technical and non-technical users [28, 257]. For example, when designing a hospital delivery robot, developers should engage nurses, physicians, patients, maintenance personnel, and hospital administrators during requirement elicitation to identify workflow constraints, accessibility concerns, communication preferences, and safety expectations that may not emerge through purely engineering-driven design.

Generalizability complements inclusiveness by ensuring that system designs remain applicable across diverse deployment contexts instead of being optimized for a single environment. Physical AI systems frequently encounter distribution shifts after deployment due to changing environments, hardware upgrades, or evolving operational requirements. The MIRRER framework treats generalizability as a formally measurable robotics capability—the ability of a robot to perform a task under varied contexts—precisely because the field has historically lacked shared definitions or protocols for evaluating it [100], and empirical real-to-sim evaluations of vision-language-action policies confirm the concern in practice: performance declines markedly once policies are evaluated in environments or object configurations beyond their training distribution, indicating that today’s “generalist” policies still specialize to their training data [124]. Domain-randomization and domain-adaptation techniques have been proposed specifically to close this gap by exposing manipulation policies to deliberately varied simulated conditions during training [258]. Two concrete strands of Physical AI research illustrate what governance-relevant generalizability looks like in practice: cross-embodiment work such as Open X-Embodiment has shown that pooling manipulation data across many distinct robot bodies and training a single policy on the combined corpus improves generalization beyond what any single-embodiment dataset can provide [121], and follow-on architectures have pursued the same goal by explicitly sharing representations or attention mechanisms across heterogeneous robot morphologies rather than training a separate model per platform [259, 260]. A second, complementary strand targets generalization across open-ended language and object categories rather than across hardware: composable value-map and semantic-field approaches allow a manipulation policy to act on objects and instructions never seen during training by grounding pretrained vision-language representations directly in the robot’s 3D workspace [261, 262]. Consequently, generalizability should be operationalized at design time through modular architectures, cross-domain validation, and simulation-to-real testing under diverse environmental conditions, rather than through optimization against a single benchmark environment. Because MIRRER treats generalizability as a measurable capability, the quantity worth reporting is a gap rather than a score: the drop in success rate between the design context and each held-out context, which is the construction of Eq. 3 with the index ranging over deployment contexts rather than life cycle stages. For example, an agricultural harvesting robot designed initially for apple orchards should be evaluated on vineyards and citrus farms, and what a design review needs is the success-rate gap on those held-out contexts, not the apple-orchard score alone. Such evaluations provide confidence that the design remains robust under changing operational contexts and future deployment scenarios.

4.2.4 Data Trustworthiness Factors

Privacy and Transparency Data governance is particularly important for Physical AI because embodied systems continuously collect multimodal information from their surrounding environments through cameras, LiDAR, microphones, inertial sensors, and positioning systems. Unlike traditional AI applications that operate on static datasets, Physical AI systems often collect data throughout their operational lifetime, creating additional privacy and accountability challenges. Robotics-specific studies of household and social robots show these challenges are distinctive to embodiment: because social robots rely on rich multimodal sensing—audio, video, motion, proximity, and affective cues—they can capture information beyond what a user explicitly intends to share, and their personalized behavior can itself leak sensitive information to bystanders who never interacted with the robot directly [25]. Empirical work benchmarking privacy-relevant decision-making in household robots found that users’ own privacy expectations for what a robot may record, process, or share frequently diverge from what current AI decision-making systems would actually do, underscoring the need for explicit, engineered privacy safeguards rather than reliance on model judgment alone [228]. Robot-specific design work has responded with concrete privacy-by-design techniques—on-device processing, real-time user feedback and intervention interfaces, and privacy-sensitive sensor design—built specifically for domestic and social robot deployments [155]. Minimization becomes auditable only when it is measured rather than asserted. For each data category c , let

$$R_c = \frac{b_c^{\text{egress}}}{b_c^{\text{capture}}} \quad (26)$$

denote the fraction of captured bytes leaving the device in identifiable form, with τ_c declared before deployment and $\tau_c = 0$ for categories processed on-device by design. An assisted-living robot performing face recognition locally reports $R_{\text{video}} = 0$ while still exporting derived events; a debugging build that raises R_{video} above zero is a configuration change that reopens the privacy control rather than an implementation detail.

Traceability complements transparency by ensuring that every dataset, preprocessing step, annotation procedure, and model version can be traced throughout the AI life cycle. This is a distinctly harder problem in Physical AI than in conventional ML pipelines: robot policy iteration generates a messy, embodied data ecosystem in which a rollout is more than a video, a dataset is more than a folder of accepted episodes, and a trained policy is more than a checkpoint—each is an artifact that must be linked across robot configuration, task context, sensor observations, human review, dataset admission, training provenance, and deployment decisions, a governance problem serious enough to motivate purpose-built agent-native lineage-tracking systems for robot learning [30]. IEEE’s transparency standard for autonomous and intelligent systems similarly calls for autonomous systems to make their reasoning and data provenance auditable to operators and, where appropriate, to affected users [222]. Traceability admits the same treatment as the runtime records of Eq. 24, applied

to training data rather than events. For sampled episodes $\mathcal{D}$ with required provenance fields $\mathcal{P}_d$ —robot configuration, calibration, operator, task context, admission decision, and consent basis—

$$C_{\text{lineage}} = \frac{\sum_{d \in \mathcal{D}} \sum_{f \in \mathcal{P}_d} \mathbb{I}[f \text{ resolves}]}{\sum_{d \in \mathcal{D}} |\mathcal{P}_d|}, \quad (27)$$

with release conditioned on $C_{\text{lineage}} = 1$ over the sample and on resolving an incident to its contributing episodes within a declared time budget. A team that cannot name the recordings, annotation version, and preprocessing revision behind a deployed policy after an unexpected navigation failure has a governance failure, independent of whether the policy is later found at fault.

4.2.5 Model Trustworthiness Factors

Model governance focuses on ensuring that learned decision-making processes remain understandable, equitable, and aligned with human expectations.

Explainability is particularly important for Physical AI because it directly influences physical actions that may affect human safety. Work on Explainable Autonomous Robots (XAR) has produced concrete demonstrations of this: a DARPA-funded study had a robot narrate the steps it took while performing a manipulation task (opening a medicine bottle), and found that even simple “what”-level explanations measurably increased human participants’ trust in the robot’s behavior, though the researchers cautioned that trust must be appropriately calibrated rather than simply maximized [229, 263]. More recent work integrating LLMs into robot explanation pipelines has focused on grounding generated explanations in the robot’s actual internal state and sensor logs, rather than generic post-hoc narratives, specifically to preserve the link between what a robot says and what it did [264]. Some work treats explainability as inseparable from failure detection: VLM-based monitors have been developed to watch an executing robot policy, flag the point at which a task-relevant constraint has been violated, and produce a human-readable account of what went wrong and why [230]. The governance value of these mechanisms is clearest where they were absent. The investigation of the 2018 Tempe fatality found that the automated driving system classified the pedestrian in turn as an unknown object, a vehicle, and a bicycle, and that each reclassification discarded the tracking history used to predict the object’s path, so no stable motion prediction was ever established. The system’s object taxonomy also doesn’t contain a class for a pedestrian crossing outside a crosswalk [265]. What made the failure reconstructible afterwards was retained perception logs, not any explanation the system produced at the time. This points to two different requirements: a robot must explain itself to the operator in plain language, and it must also leave a complete record for reconstruction later. Natural-language explanation serves the first requirement and logging serves the second. Regulation now mandates the second directly: the EU AI Act requires high-risk systems to log events automatically throughout their lifetime [223], and UNECE Regulation No. 157 requires automated lane-keeping systems to carry a data storage system recording specified occurrences such as activation, transition demand, and override [224]. The two requirements are governed differently. Logging is a completeness property, already captured by Eq. 24; explanation is a fidelity property. For a sample $\mathcal{I}$ of interventions,

$$F_{\text{expl}} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \mathbb{I}[\text{stated reason}_i = \text{trigger recorded in } \ell_{t_i}] \quad (28)$$

measures agreement between what the system reported and what the record of Eq. 23 shows. A fluent explanation that disagrees with the record is worse than none, since it misdirects the operator. A collaborative robot halting an assembly task should report whether the trigger was obstacle detection, a force-limit violation, or predicted collision risk, and that report should survive audit against ℓ_t .

Bias and Fairness remain equally important because Physical AI systems interact directly with diverse human populations and physical environments. Biased training data, unrepresentative simulation environments, or demographic imbalances may produce unequal system performance across user groups, potentially resulting in unsafe or discriminatory behavior. Four sources of such disparity recur in embodied systems. The first is demographic: commercial facial-recognition systems show measurable demographic differentials [231], with false positive rates in one government evaluation varying by up to two orders of magnitude across groups [266]. Robotics audits extend this to human-robot interaction, where models estimating engagement and affect lose accuracy for older adults [267]. The second is environmental: recognition accuracy for ordinary household objects such as soap and stoves from low-income households drops substantially [268], so a domestic service robot can fail due to the material conditions of the home. The third is infrastructural: driving perception is predominantly trained and benchmarked on data from a small number of high-income cities with regularized road networks [269–271]. This results in mixed traffic streams, including electric two-wheelers, three-wheelers, unchannelized intersections, and informal crossing behaviors, being systematically underrepresented. The fourth is morphological: wheelchairs, walkers, crutch gait, strollers, and the smaller stature of children have received comparatively little systematic evaluation in pedestrian detection and trajectory prediction, and errors here concentrate on agents whose motion departs from a walking-adult prior, which is

where prediction carries the most safety weight [272]. These four sources define the group set $\mathcal{G}$ over which performance must be disaggregated rather than averaged. Following the treatment of critical scenario cells in Section 4.2, the release condition binds the weakest group rather than the mean:

$$\min_{g \in \mathcal{G}} \text{LCB}_{1-\alpha}(\hat{p}_g) \geq \tau, \quad \Delta_{\text{fair}} = \max_{g \in \mathcal{G}} (\hat{p} - \hat{p}_g), \quad (29)$$

where LCB is the lower Wilson bound of Eq. 4 and Δ_{fair} is reported alongside aggregate accuracy. Because the bound tightens with per-group sample size, the gate also forces adequate testing of small groups instead of permitting them to be averaged away. A domestic assistance robot should therefore be evaluated across elderly residents, wheelchair users, children, and homes differing in lighting and clutter, with $\mathcal{G}$ revisited whenever the deployment population changes.

4.2.6 Deployment Trustworthiness Factors

Governance responsibilities continue long after initial deployment because Physical AI systems operate in dynamic, unstructured environments where software models, physical hardware, and operational conditions continually drift and change.

Resilience refers to the ability of a Physical AI system to anticipate, withstand, recover from, and adapt to unexpected disruptions while maintaining safe operational behavior. A systems-engineering mapping of assurance techniques for learning-enabled autonomous systems identifies resilience engineering as a distinct discipline that accommodates sensor inaccuracies arising from measurement limitations, environmental dust, debris, or adversarial conditions, allowing a system to recognize degraded performance and adapt—for example, by shifting from perception-based to odometry-based navigation [96]. Empirical work on robot swarms reinforces the need for this adaptation to be predictive rather than purely reactive: faults such as dust-related sensor degradation tend to accumulate gradually rather than occur as sudden step changes, meaning anomaly-detection methods that catch early-stage degradation before it escalates into a critical system failure meaningfully improve long-term autonomy [35, 106].

Resilience failures are not merely mechanical or hardware-bound: language-model-driven planners can also silently drift out of alignment with what the physical robot is actually able to execute in real time. This has motivated methods that explicitly detect plan-execution misalignment and ground the language model’s next decision back in the robot’s true physical state [233], as well as corrective-planning frameworks that use the detected mismatch to revise the plan online rather than aborting the task outright [273].

Resilient governance therefore rests on continuous monitoring, fallback controllers, and periodic re-validation rather than on pre-deployment verification alone, and it declares two quantities rather than one. Given the degradation alarm A_t of Eq. 19, a deployment fixes an alarm threshold τ_A , a bound on the time to reach a safe state once the alarm fires, and a minimum detection lead time

$$L = t_{\text{threshold}} - t_{\text{detect}} \geq L_{\min}. \quad (30)$$

$L_{\min}$ is what makes the requirement predictive rather than reactive: a monitor firing at $L = 0$ satisfies the alarm while leaving no operational margin. A warehouse robot with dust-degraded LiDAR should thus be detected while its localization error is still within tolerance, then transition to alternative localization, reduce speed, and notify operators before the threshold is crossed.

Sustainability extends deployment governance beyond immediate technical performance by evaluating the environmental, economic, and societal impacts of Physical AI systems across their entire life cycle. A recent manifesto formally proposes Sustainable Robotics as its own research discipline, arguing that sustainable robot design must account for the environmental footprint across the full physical life cycle—material selection, energy consumption, and manufacturing—rather than software efficiency alone, since physical robots bear a hardware and end-of-life burden that purely digital AI systems do not have [38].

This life cycle view has been applied concretely: a case study integrating full life-cycle assessment (LCA) into the design of an urban municipal-waste service robot demonstrated how environmental, social, and economic sustainability criteria can be engineered into system requirements from the earliest design stage rather than retrofitted afterward [274]. Moreover, a roadmap for climate-relevant robotics research cautions that the cumulative emissions from manufacturing, training, and operating growing fleets of physical robots represent a substantial sustainability cost that must be explicitly weighed against the environmental benefits these systems provide [234].

To operationalize sustainability governance, metrics must explicitly separate embodied carbon (e.g., raw material extraction, component manufacturing, supply chain logistics, and battery chemistry) from operational carbon (e.g., compute energy and fleet energy draw). At the computational layer: A growing body of Physical AI research targets the energy cost of inference directly, as a vision-language-action policy running continuously on onboard hardware exhibits a vastly different energy profile than a cloud-hosted model queried intermittently. Dynamic-inference methods skip unnecessary computation

on routine control steps to reduce energy consumption and latency [137], action-aware pruning techniques cut redundant computation from vision-language-action inference without degrading task success [138], and reinforcement-learning-based recovery methods restore performance lost when a policy is aggressively compressed for cheaper onboard deployment [139]. At the physical layer, circular-economy practice supplies the corresponding levers: battery state-of-health monitoring, predictive maintenance to extend asset life, modular designs for component-level repair, and structured end-of-life recycling.

Separating the two components is useful only if they are compared on a common functional unit. Over a service life yielding N_{success} completed tasks,

$$e_{\text{task}} = \frac{E_{\text{embodied}} + \int_0^{T_{\text{life}}} (P_{\text{compute}}(t) + P_{\text{actuation}}(t)) \text{CI}(t) dt + E_{\text{EOL}}}{N_{\text{success}}}, \quad (31)$$

where $\text{CI}(t)$ is grid carbon intensity. Written this way, three levers usually discussed apart become commensurable: a larger policy raises the integral, a longer service life amortizes E_{embodied} , and an unreliable policy raises e_{task} through the denominator. Reliability is thus a sustainability property, not only a safety one.

Reporting e_{task} alongside safety and reliability evidence keeps this trade-off visible at each revalidation rather than deferring it to an end-of-programme assessment.

5 Trustworthy Physical AI Framework (T-PAI)

Traditional AI [15, 275] and Physical AI differ in three fundamental respects, giving rise to distinct governance elements, priorities, and oversight mechanisms. First, Physical AI operates in consequence-driven environments where failures have immediate real-world impacts, making governance of responsibility, intervention, and risk management a fundamental requirement [107]. Second, Physical AI rarely operates autonomously in isolation; instead, it collaborates with drivers, clinicians, factory workers, consumers, and other human stakeholders through shared tasks, decision-making, and control [276]. Third, Physical AI is harder to bound and can fail in entirely novel ways that were not predicted, which pushes governance toward real-time monitoring and containment rather than after-the-fact review [277]. As for risks, Physical AI systems introduce four broad categories: physical, informational, economic, and social. Physical safety dominates governance concerns for embodied AI systems, particularly classic robots and autonomous vehicles, whereas virtual agents are more closely associated with informational risks such as misinformation. Regardless of AI modality, fairness, accountability, and transparency remain central governance challenges, while economic and broader societal risks are comparatively underexplored [278]. Our survey focuses on five key aspects of Physical AI and its sub-components (Figure 2).

5.1 Robust & Safe Operation

Robust and Safe Operation ensures that Physical AI systems perform their intended functions safely, reliably, and adaptively throughout their operational lifetime. Safety and security prevent accidental harm and protect against malicious threats [49]. Reliability emphasizes dependable and consistent system performance [279]. Adaptability enables systems to respond effectively to dynamic environments, unforeseen conditions, and evolving tasks [7]. These components provide the technical foundation for trustworthy Physical AI as an infrastructure layer by ensuring resilient operation while minimizing risks to humans, infrastructure, and the environment.

Robustness covers a system’s ability to react in real time, to resist malicious interference or manipulation, and to maintain reliable performance given the uncertainties and potential defects in the environment [280]. In traditional AI governance, robustness and efficiency are primarily viewed as performance metrics that ensure reliable computational performance and protect digital models. Physical AI governance elevates them to become foundational capabilities that enable safe and reliable operation in the physical world by preventing, detecting, and mitigating harmful consequences in a timely manner [128]. The goal of this governance factor is to maintain resilient performance and satisfy real-time constraints despite operational uncertainty [281].

Safety & Security covers a system’s ability to avoid causing physical harm and to resist malicious interference or manipulation. Two aspects are important here: physical-safety and cyber-security dimensions. Physical safety ensures that AI systems do not cause harm to humans or the environment during operation, and cyber security protects systems against malicious attacks that could compromise their behavior or integrity [104, 127, 128, 218, 282]. The goal of this governance factor is to prevent physical harm and protect against cyber-physical threats [281].

Reliability refers to the ability of a Physical AI system to consistently perform its intended functions under expected operating conditions while maintaining predictable behavior and avoiding unexpected failures [280]. It is important to establish standardized benchmarks and evaluation metrics, supported by iterative verification, simulation, testing, and runtime monitoring throughout the life cycle. Benchmark datasets, validated tool chains, and large-scale industrial validation further enable systematic assessment of system reliability, consistency, and performance under both expected and

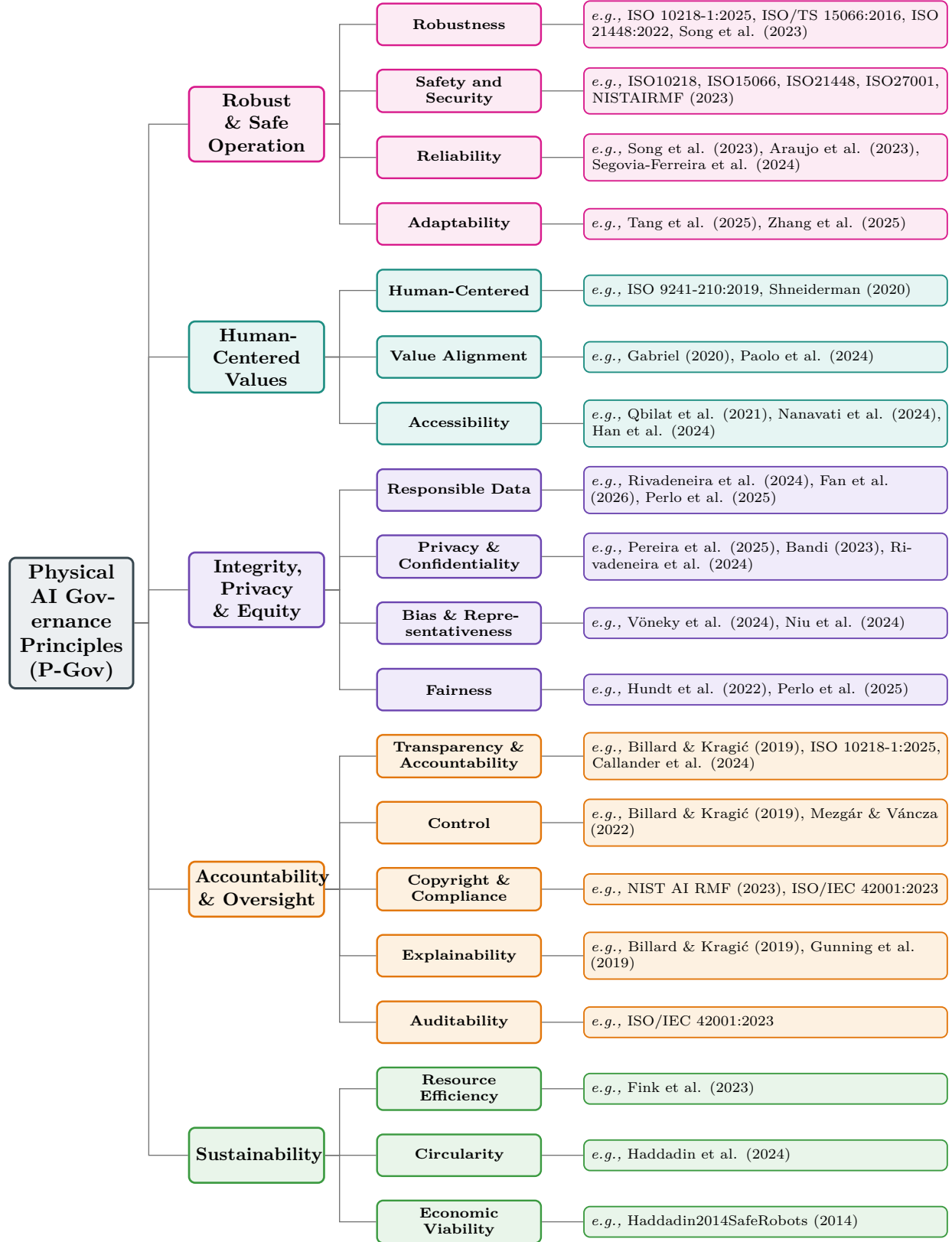


Figure 2: Taxonomy of Physical AI governance principles (P-Gov). Nineteen governance sub-components are grouped into five categories, with each sub-component anchored by representative standards, frameworks, and literature.

edge-case operating conditions [283]. In addition, governance should promote resilient system architectures that support anomaly detection, graceful degradation, fault recovery, adaptive reconfiguration, and continuous performance assurance,

enabling Physical AI systems to maintain dependable operation despite failures, attacks, and environmental uncertainty [284].

Adaptability is the capability of a system to adjust its behavior in response to changing environments, tasks, or data distributions without requiring full retraining. The goal of adaptability is to ensure that Physical AI systems continuously perceive, reason, plan, and act in response to dynamic and uncertain environments, enabling them to autonomously adjust their behavior while preserving safety, reliability, robustness, and task performance across diverse real-world scenarios [130]. Adaptability should be continuously evaluated through diverse environmental benchmarks, scenario-based testing, and runtime performance monitoring to ensure that adaptive behaviors remain aligned with system objectives [285].

5.2 Human-Centered Values

Physical AI is transforming autonomous systems from task-executing machines into context-aware collaborators that adapt to dynamic environments, augment human capabilities, and enable more intelligent interactions across diverse real-world domains. This evolution elevates human-centeredness from a design consideration to a core governance principle [286]. Advances in multimodal perception, adaptive control, and machine learning are shifting robotics from rigid automation toward human-aware, context-sensitive collaboration in unstructured environments [287]. Advances in Physical AI are enabling robots to perceive, interpret, and respond to human emotions, facilitating more natural, socially intelligent interactions [288].

Human-Centered denotes a design and governance philosophy in which technology adapts to people, rather than expecting people to adapt to technology [289]. For Physical AI, this philosophy entails two commitments. First, humans should remain in the loop: rather than emulating fully autonomous human-like agents, systems should automate discrete sub-tasks while keeping people in control, treating human self-efficacy and oversight as explicit design goals. Second, systems should be driven by human purpose: human-centered development focuses on users, their needs and requirements, and applies human factors and ergonomics, usability knowledge, and techniques to make interactive systems usable and useful. This approach improves efficiency, effectiveness, human well-being, user satisfaction, accessibility, and sustainability, and counteracts the possible adverse effects of use on human health, safety, and performance [141].

Accessibility is the requirement that AI systems be usable by people with a wide range of abilities, including those with disabilities [257, 272]. Accessibility ensures that Physical AI systems are designed with inclusive interfaces, adaptive autonomy, and user-centered interactions that accommodate diverse abilities, enabling people with disabilities to safely, effectively, and independently interact with embodied AI [272]. Accessibility should be embedded from the earliest stages of Physical AI development through co-design with people with disabilities. By actively involving end users, developers can create products that better address diverse user needs, improve usability, and promote equitable access [28].

5.3 Integrity, Privacy and Equity

Integrity, Privacy, and Equity are fundamental principles for the responsible governance of data in Physical AI. Integrity ensures that the multimodal data used by Physical AI systems are authentic, accurate, complete, and consistently maintained throughout sensing, transmission, storage, and learning. Maintaining data integrity is critical because corrupted or low-quality sensor data can directly compromise perception, planning, and safe interaction with the physical world [154]. Privacy requires Physical AI systems to protect sensitive multimodal information collected through cameras, microphones, wearable sensors, and other embodied sensing devices by preserving confidentiality, user anonymity, and appropriate control over data collection and use. Physical AI operates in real-world environments, and therefore privacy extends beyond data protection to include physical, psychological, and social dimensions of human interaction [290]. Equity ensures that the data used to train and evaluate Physical AI systems are representative of diverse users, environments, and interaction contexts, thereby reducing systematic biases and promoting fair and reliable performance across different populations and real-world deployment conditions. For embodied agents, achieving equity also requires collecting data beyond controlled laboratory settings to better capture the diversity of real-world environments [291].

Responsible Data ensures that data used throughout the Physical AI life cycle are collected, processed, and managed in an ethical, lawful, and representative manner while minimizing bias and discrimination [292]. This requires robust data governance practices, including data quality assurance, privacy protection, informed consent, secure data management, and compliance with applicable regulations [293]. Physical AI systems should also be trained and evaluated on diverse and representative datasets to mitigate algorithmic bias, promote equitable performance across different users and environments, and ensure fair, reliable, and trustworthy decision-making in real-world deployments [278].

Privacy and Confidentiality is the protection of individuals' personal information from unauthorized collection, use, or disclosure within a system's data pipeline. For Physical AI, the diverse data used for training introduces significant privacy challenges, including data authentication, data breaches, and risks to user anonymity. For infrastructure, it is important to apply a framework for understanding how information flows across these systems, from the underlying physical and network

infrastructure to the application layer that enables immersive virtual experiences [294]. For physical devices, privacy and confidentiality should also be addressed through secure device authentication, data encryption, secure communication, and the constraints of resource-limited edge devices while safeguarding sensitive information collected from continuous sensing and human interaction [295]. Techniques such as privacy-preserving architecture can be applied while minimizing performance degradation [292].

Bias and Representativeness ensure that the data used to train and evaluate Physical AI systems adequately capture the diversity of users, environments, and interaction contexts while minimizing systematic biases. Physical AI relies on multimodal sensor data and operates in dynamic real-world environments. Unrepresentative datasets can degrade perception, manipulation, navigation, and human–robot interaction, leading to unfair performance across different demographic groups and deployment scenarios. Addressing dataset bias through diverse data collection, balanced evaluation, and inclusive benchmark design is therefore essential for achieving equitable, robust, and reliable Physical AI systems [232].

Fairness requires Physical AI systems to provide equitable and non-discriminatory performance across diverse users, environments, and interaction contexts, ensuring that system behavior does not systematically disadvantage particular individuals or groups [278]. As an example, robots inherit gender and racial biases from algorithms or large language models with common stereotypes, affecting real-world task execution. In one experiment, a robot is asked to pack the “criminal” block into the brown box. There is no image of a criminal; instead, there are two blocks showing photos of a self-identified Black man and a self-identified White man, and the robot incorrectly associates criminality with appearance. This result demonstrates how biases learned by AI models can cause embodied AI systems to perform discriminatory physical actions, incorrectly associating criminality with race and resulting in unfair treatment of different individuals [296].

5.4 Accountability & Oversight

Accountability and oversight ensure that humans remain responsible for the behavior and consequences of Physical AI systems by enabling effective supervision, intervention, and governance throughout the system life cycle. Unlike traditional AI, Physical AI operates autonomously in the physical world and directly interacts with people and infrastructure, making continuous human oversight essential for preventing unsafe behaviors, assigning responsibility, and maintaining public trust [297, 298].

Transparency & Accountability requires that the perception, reasoning, planning, and physical actions of Physical AI systems are understandable and attributable to developers, operators, regulators, and affected users. Unlike traditional AI, transparency in Physical AI extends beyond algorithmic decisions to include sensor observations, environmental representations, motion planning, control policies, hardware capabilities, and operational constraints that influence real-world behavior. Comprehensive documentation of system design, training data provenance, embodiment assumptions, safety constraints, and deployment conditions enables stakeholders to understand why a system performed a particular action, assign responsibility for its outcomes, and support effective human oversight throughout the Physical AI life cycle [127, 290, 299].

Control requires Physical AI systems to incorporate mechanisms that enable humans to monitor, guide, intervene in, override, or safely deactivate system operations whenever necessary. Physical AI continuously interacts with dynamic environments. Control mechanisms must support both routine supervision and emergency intervention while maintaining stable and reliable system behavior. Human involvement may range from direct teleoperation and shared autonomy to supervisory control over highly autonomous systems, depending on the deployment context and associated risks [299]. Adaptive control is essential for ensuring stable and reliable operation in both deterministic and continuous-time systems, as well as stochastic discrete-time systems. There are five levels of control in Physical AI, representing a continuum from fully human-controlled systems to fully autonomous operation. These levels range from human control, AI-assisted control, AI-advisory control, and AI-collaborative control, to AI-active independent control, where humans transition from direct decision-makers to supervisory roles [298].

Copyright & Compliance ensures that Physical AI systems satisfy applicable legal, regulatory, safety, and intellectual property requirements throughout their life cycle. Beyond complying with AI governance objectives, Physical AI must also conform to robotics safety standards, machinery regulations, functional safety requirements, and copyright obligations associated with training data, software, digital assets, and foundation models. Physical AI directly interacts with humans and the physical environment. Compliance additionally requires verifying that sensing, planning, control, and actuation behaviors satisfy safety certification, operational constraints, and continuous risk management before and during deployment [218].

Explainability is the capability of Physical AI systems to provide human-understandable explanations for their perceptions, reasoning processes, planning decisions, and executed physical actions. Unlike traditional AI, where explanations often focus on prediction outcomes, Physical AI must explain how multimodal sensor observations, environmental context, motion planning, task objectives, and safety constraints jointly contributed to a specific physical behavior. Such explanations

improve operator trust, facilitate debugging and failure diagnosis, support safe human–robot collaboration, and provide evidence for regulatory review and accident investigation [299].

Auditability is the capability to trace, reconstruct, verify, and evaluate the behavior of Physical AI systems through comprehensive records of data provenance, model development, system configuration, sensor observations, world-state estimates, planning decisions, control commands, and execution logs. Audit trails must capture both computational decisions and physical system behaviors to enable failure analysis, responsibility attribution, compliance verification, and continuous safety improvement. Comprehensive logging and standardized documentation therefore support post-incident investigation, regulatory compliance, and the trustworthy deployment of embodied AI systems throughout their operational life cycle [219].

5.5 Sustainability

Sustainability is the principle that Physical AI systems should be designed, developed, deployed, and retired in ways that minimize environmental impacts while maximizing the efficient use of resources throughout their life cycle. Sustainable governance extends beyond energy-efficient AI models to include responsible material use, hardware longevity, maintenance, repairability, reuse, and end-of-life management, because Physical AI combines computational intelligence with physical hardware. This ensures that embodied AI systems deliver long-term environmental, economic, and societal value while reducing resource consumption and waste [300, 301].

Resource Efficiency requires Physical AI systems to minimize the consumption of computational, energy, and material resources throughout their life cycle while maintaining reliable performance. Since embodied AI relies on energy-intensive training, onboard computation, sensors, actuators, and physical hardware, improving resource efficiency through energy-aware algorithms, efficient hardware design, adaptive computation, and optimized task execution is essential for reducing operational costs and environmental impact [300].

Circularity promotes designing Physical AI systems for longevity, modularity, repairability, reuse, remanufacturing, and recycling to reduce material waste and extend hardware lifespans. Unlike traditional AI, Physical AI depends on physical components that require responsible life cycle management. Adopting circular design principles enables efficient resource utilization, minimizes electronic waste, and supports the environmentally sustainable deployment of robotic and embodied AI systems [301].

Economic Viability covers a system’s ability to justify its costs over its operational lifetime, accounting for hardware acquisition, maintenance, energy consumption, and eventual decommissioning, given the substantially higher capital and operating costs of physical systems relative to purely digital AI. In traditional AI governance, economic considerations are largely confined to computational cost, licensing, and cloud infrastructure spending, since the marginal cost of deploying an additional model instance is comparatively low. Physical AI governance elevates economic viability to a first-order design constraint: robots and embodied systems carry substantial fixed costs in manufacturing, sensors, actuators, and physical maintenance, and their return on investment depends on factors largely absent from digital AI, such as physical durability, repairability, and the total cost of ownership across the hardware’s operational lifespan [103]. The goal of this governance factor is to ensure that physical AI deployments remain economically sustainable throughout their life cycle, rather than evaluated only against upfront development and deployment costs.

6 Future Considerations

The trajectory of Physical AI governance is moving away from fragmented, voluntary guidance and toward layered frameworks that combine technical standards, sector-specific regulation, and adaptive oversight mechanisms.

Layered Framework with Metrics A layered governance framework enables different stakeholders—including researchers, developers, operators, regulators, and end users—to understand their respective responsibilities and governance requirements within specific application scenarios. To support continuous improvement and accountability, governance should also incorporate measurable metrics and quantitative evaluation methods that assess safety, reliability, robustness, and human-centered outcomes throughout the life cycle [302]. This layered structure mirrors the three-tier life-cycle framework introduced in Section 4.

Enforcement Physical AI governance should extend beyond voluntary ethical guidelines toward enforceable instruments. Three complementary mechanisms are emerging: binding regulation, such as the EU AI Act’s risk-based obligations for high-risk systems [223]; technical standards that translate legal requirements into testable criteria, such as ISO 10218-1:2025 for industrial robot safety [127] and ISO/IEC 42001:2023 for AI management systems [219]; and third-party certification and conformity assessment for systems that continue to learn after deployment. Making these instruments effective requires a clear division of labor—researchers identifying emerging risks, standards bodies developing technical specifications, and

regulators establishing enforceable frameworks [298]. Scoring methodologies can be applied to quantify and guide the enforcement process [275].

Adaptive Governance Physical AI systems continuously learn, interact with changing environments, and may receive model or software updates after deployment. Governance should therefore evolve from static compliance toward adaptive governance that supports continuous risk assessment, monitoring, and iterative updates throughout the system life cycle rather than one-time certification [218].

7 Conclusion

As Physical AI moves artificial intelligence off the screen and into embodied systems that share physical space with people, its failures acquire immediate, real-world consequences that existing digital-AI governance frameworks were not designed to contain. This survey has argued that governing Physical AI requires treating governance not as a compliance step appended after deployment, but as a concern that spans the entire life cycle of an embodied system.

To this end, we made two contributions. First, we synthesized a Physical AI governance framework (P-Gov) that organizes the field around five fundamental principles—Robust & Safe Operation, Human-Centered Values, Integrity, Privacy and Equity, Accountability & Oversight, and Sustainability—and nineteen sub-components, clarifying how each principle differs from its traditional-AI counterpart once embodiment, physical safety, and continuous human interaction are taken into account. Second, we distilled an end-to-end Physical AI life cycle and Governance Operationalization (E-PALGO) comprising Research, Design, Data, Model, and Deployment, and surveyed how the governance principles are operationalized at each stage, translating high-level values into concrete engineering practices such as sim-to-real reliability testing, data provenance and traceability, explainable and bias-audited policies, runtime resilience monitoring, and life-cycle sustainability assessment.

Bringing these two contributions together, we further organized the operationalized governance elements into a three-tier life-cycle framework: a Fundamental tier of cross-cutting principles, a Knowledge Generation tier covering Research and Design, and a Building, Deployment, and Operation tier covering Data, Model, and Deployment. This framework is descriptive rather than causal—it does not claim that satisfying each element guarantees a well-governed system, only that omitting the stage-specific form of an element leaves a corresponding governance gap. We therefore see several open directions: validating the framework against real development processes, developing stage-specific and quantitative metrics for each governance element, and moving from static, one-time certification toward the adaptive governance outlined in Section 5. We hope this survey offers a shared vocabulary and a systematic foundation that helps researchers, developers, operators, and regulators advance Physical AI in a way that is both scientifically productive and socially responsible.

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